

DEVELOPMENT AND CONTROL OF A MODULAR ARTIFICIAL SHOULDER PROSTHETIC DEVICE

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Declaration

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Chapter 1 - Introduction

People who suffer severe injuries to their arm / suffer from ailments that tend to damage the arm such as bone cancers, may have to undergo amputations of the damaged arm. If the amputation is a shoulder disarticulation or forequarter amputation, the amputee will lose his / her shoulder as well as the collarbone. For such an amputee, a shoulder prosthetic device is required to recover the usability of the arm, at least to a minimal level. Hence a need arises to develop a prosthetic device for shoulder amputees who had to undergo forequarter amputation and shoulder disarticulation, to facilitate the use of a prosthetic elbow, lower arm and wrist [1–3]. Furthermore, the control of the shoulder prosthetic device should be facilitated as much as possible to the subject, hence an electromyography (EMG) based control system is implemented to the shoulder prosthetic device [4].

1.1 Context

Accidents that occur under various situations (traffic accidents, workplace accidents, damage from ammunition) sometimes lead to the amputation of a victim's arm from the shoulder, resulting in the comprehensive loss of use from the arm [5]. Under such a circumstance, the victim will face many difficulties in carrying out tasks on a day to day basis. To assist such disabled individuals, prosthetic devices have been proposed and developed. However, most of them are yet to be perfected. No commercially cost effective active shoulder prosthetic system is available for purchase at present. Most of the available systems have been developed in international research laboratories in United States, United Kingdom and other European countries [5]. Even if such a system becomes available, the cost would be prohibitively high, in order to implement in Sri Lankan context. At present, no such prosthetic device has been developed in Sri Lanka as well. In order to fulfil the necessity for amputees our group has initiated the development of modular artificial prosthetic device controls using electromyography (EMG) signals [4].

1.2 Research gap

Extensive research has been conducted on developing active shoulder prosthetic device with 3 DOF, to be used alongside prosthetic elbow, lower arm and wrist with terminal device. However no such research has been conducted in Sri Lanka in order to develop a shoulder prosthetic device with 3 DOF and EMG based control. Research had been carried out international re-

search laboratories to develop such a shoulder prosthetic device, either as a modular unit or as a part of combined prosthetic arm. However those developed artificial prosthetic devices are not financially feasible for Sri Lankan context. Therefore, an amputee in Sri Lanka may not be able to utilise such a device. In addition, most prosthetic devices are developed with a particular subject (amputee) in mind. Hence a need of a modular prosthetic unit arises in order to cater to a wide variety of victims. When developing the prosthetic device, attention must be paid to develop a comfortable socket for the user as well. In order to achieve these goals, this project is intended to develop this modular artificial shoulder prosthetic device.

1.3 Aim

- Develop a modular prosthetic device which can replicate the motion of the shoulder, and can be controlled by the electromyography (EMG) signals.

1.4 Objectives

- To design and develop a 3 DOF modular shoulder prosthetic device.
- To evaluate and validate the effectiveness of motion generation of the prosthetic device.
- To develop an EMG signals - based method to robustly control the prosthetic device.
- To evaluate the effectiveness of the proposed control method with the shoulder prosthetic device by carrying out experiments.

1.5 Methodology

- Research on existing solutions for modular shoulder prosthetic device using a comprehensive literature review.
- Carryout a conceptual design, for modular shoulder prosthetic device with 3 degree of freedom, and a suitable shoulder mounting socket.
- Study about EMG signal acquisitions and conditioning.
- Develop 3D CAD model of the shoulder prosthetic device and shoulder mounting socket, and conduct a stress analysis.
- Fabrication of the shoulder prosthetic robot and shoulder socket.
- Design and implementation of the EMG.
- Validate/evaluate the developed system using experiments.

Chapter 2 - Literature Review

In this literature review, the relevant literature is categorised into three main sections and reviewed separately. First section is an introductory section which provides an insight to upper limb amputations including statistical data and prosthetic solutions available for shoulder amputees. Second section focuses on hardware construction involved in active shoulder prostheses. The third section is regarding control of active prostheses, with attention to electromyographic signal based control.

2.1 Biomechanics of Shoulder joint

2.1.1 Shoulder Anatomy

The shoulder is the joint which connects the upper limb to the trunk in the human body. The shoulder joint is however, a complex collection of bones, muscles, ligaments, and is more appropriately referred to as the shoulder girdle [6]. The shoulder girdle, which is also referred to as the pectoral girdle, allows an exceptional freedom of movement for the upper limb [6, 7]. The bones and muscles which make up the shoulder girdle are;

- Bones - Scapula, Clavicle
- Muscles - Deltoid, Pectoralis Major, Trapezius [7]

The shoulder girdle comprises of three joints, two of which are depicted in Figure 2.1 [6]:

- Glenohumeral (GH) joint
- Acromioclavicular (AC) joint, Trapezius
- Sternoclavicular joint [6]

2.1.2 Upper Limb Amputations

The upper limb is the collective reference for the shoulder girdle, the arm, forearm and the hand [8]. Amputation is described as a surgical procedure for partial or comprehensive removal of a limb [9]. Incidence of amputations is at 1.5 per every 1000 people, and an estimated 10 million amputees are in the world [10]. Out of this, an estimated 3% - 15% had been subject to amputations involving upper limb [9]. Amputations involving the upper limb can

be required due to various reasons, such as war – related injuries and other accidents (traffic related, workplace – related, household) which results in trauma, conditions present when born (congenital), and diseases.

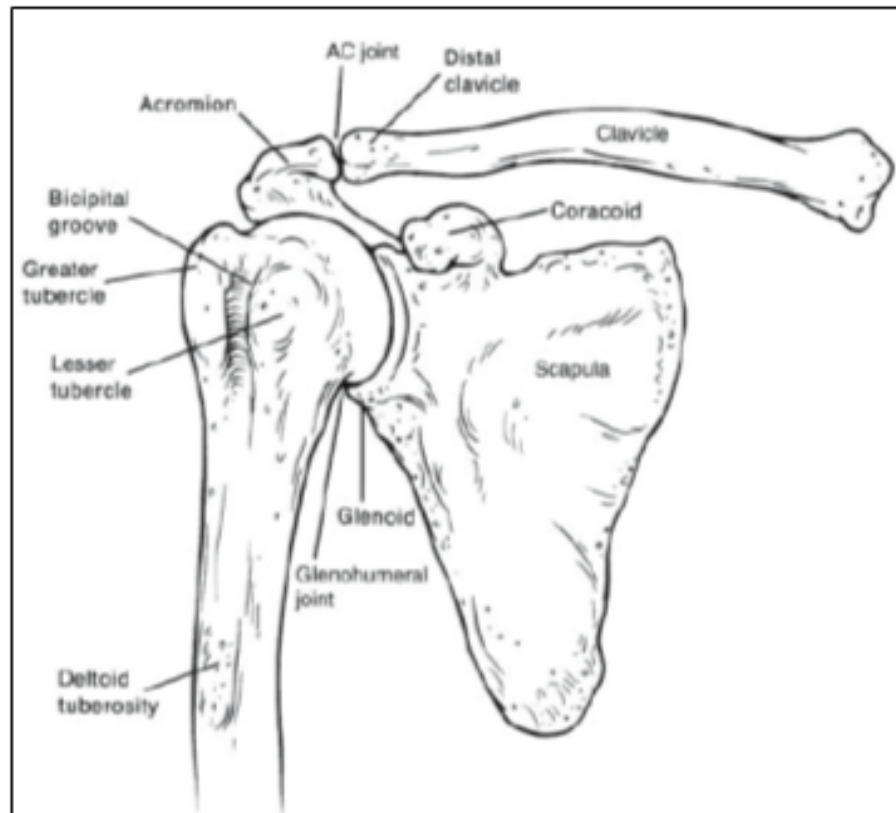


Figure 2.1. Anterior view of bones in the shoulder

Table 2.1. Summary of upper limb amputee related statistics.

Source	Study conducted by	Relevant country	Amputation type	Cause of amputation	Percentage out of
[4]	Not stated	USA	Upper limb	Trauma	75% of acquired upper limb amputations of males between 15 - 45 years of age
[4]	Dillingham	-do-	Upper limb	Trauma	70% of upper limb amputations (1988 - 1996)
[4]	Keljaa	Denmark	Upper limb	Overall	3% of all amputations (1992)
[4]	-do-	-do-	-do-	Trauma	74% of all upper limb amputations

* The study found out that majority of trauma related amputations occurred due to accidents occurring in industrial jobs, either skilled or unskilled, and traffic accidents [11].

Based on the damaged part of the arm, different levels of amputations may be required to protect the rest of the body from being subject to the ailment (such as cancers) in the case of disease related amputations, or to remove organs which have become dysfunctional owing to the injuries sustained, in case of trauma – related amputations [9]. The types of upper limb amputations are

- Digit – Only portion of a finger / multiple fingers are amputated
- Partial hand – A portion of the hand is amputated
- Wrist disarticulation – The entire hand has to be amputated from the wrist
- Transradial – The entire hand, along with a portion of the forearm, below elbow, has to be amputated
- Elbow disarticulation – The forearm is amputated from the elbow joint
- Transhumeral – The arm is amputated above the elbow joint, below the shoulder joint
- Shoulder disarticulation – When less than 30% of humeral stump remains after the amputation, it is called a shoulder disarticulation.
- Forequarter amputation – If the entire upper extremity, clavicle and scapula are removed during amputation, it is called a forequarter amputation, and leads to complete loss of use from the upper limb subject to amputation [12–14]. Forequarter amputation, which is referred as interscapulothoracic amputation in medical terms, had largely been caused by amputations due to high grade bone sarcomas involving proximal humours and scapula [13]. Shoulder disarticulation and forequarter amputation stages are depicted in Figure 2.2 [14]. The first figure (from left) depicts humeral neck remnant, the second depicts shoulder disarticulation and the third depicts forequarter amputation level.



Figure 2.2. Clinical representation of the limb which remains after amputation

2.1.3 Types of prostheses available for shoulder amputees

When an amputee has to undergo shoulder disarticulation or forequarter amputation, he / she will face complete loss of use from the upper limb subject to amputation. Hence, there is a need for the amputee to find a method to minimize the consequences of the loss of arm. Usually the contralateral upper limb (the remaining upper limb) will take over most of the functionality lost from the amputated limb, however there is a limitation to the ability to regain the functionality as there are many day to day tasks which require both hands [9]. Shoulder amputees have a choice of selecting between the three options of not using a prosthetic device, using a cosmetic (passive) prosthetic device or an active prosthetic device. Passive prostheses are used for cosmetic purposes only, and cannot be actively used for day to day activities. Reduced weight, energy and cognitive requirements and better appearance are some of the key reasons why amputees tend to resort to passive prostheses. Hence it is understood that the design and development of active prostheses should consider improvements in weight reduction and improvement of appearance. Active prostheses can be further categorized as body powered, externally powered and hybrid systems. Body powered shoulder prostheses can present a challenge for the user, and can result in discomfort when used for prolonged periods [14]. Active and passive prostheses are further discussed in chapter two of this review. In addition to these two main types of prostheses, activity specific prostheses are available as well. However this type is not discussed since the project concerned here is intended to develop a general utility artificial prosthetic shoulder. When developing the prosthetic solution for amputees, it is important to focus on the weight of the prosthesis, the accumulation of heat in the socket, the stability of the device, and the ability for self – donning [14].

2.2 Hardware Construction of Active Shoulder Prostheses

2.2.1 Introduction

Hardware design and constructions is one of the important aspect of this project as the weight and the arrangement of the prosthesis is crucial. This section consist of existing hardware constructions designs, components and hardware capabilities of available active shoulder prosthesis designs.

2.2.2 Types of Prostheses

- Passive prostheses
- Active prostheses

2.2.3 Passive prostheses

Passive prostheses or cosmetic prosthesis has limited functionalities. Visual appearance the major concern with no controlling mechanisms. There are variety of colourful passive prostheses available in the market. Advantages of passive prostheses are less weight, high cosmetic appearance, reduced energy and cognitive requirements, because active prostheses are still in research stage. The repair costs are relatively less and easy to maintain compare with active prostheses [14, 15].

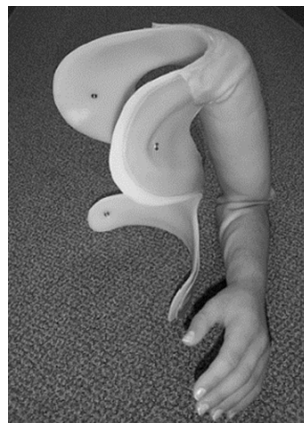


Figure 2.3. Passive prosthesis arm with shoulder joint

2.2.4 Active prostheses

Subject can control the active prostheses. Tasks such as grasp, manoeuvre objects can be done using active prostheses. Active prostheses belongs to more advanced prostheses category. There are two different design types of prostheses called body powered and externally powered prostheses [16].

Body powered prostheses are technologically less complex, compared to externally powered prostheses. Body powered prostheses are mechanically connected to the user's body, with a control interface such as cables. Body powered prostheses can be used even in harsh, dirty environment conditions. This type of prostheses can be actively controlled by cables or some mechanical method. Therefore the controllability requires generation of forces through the rest of the body in areas such as torso, and this can result in discomfort for the user. The advantage is the durability, elimination of need of electric power supply and the ability for user to repair for a lesser cost than externally (electrically) powered prostheses. Prosthesis arm with a hook is an example for this category of prostheses [14, 16, 17].



Figure 2.4. Active body powered prosthesis arm with shoulder joint

Hybrid prostheses combine the control methods of both body powered and externally powered prostheses. Hence hybrid prostheses can possess the advantages in both methods and reduce individual disadvantages. Ability to provide fast flexion/ extension is a key advantage of hybrid method. Hybrid prostheses attempt to minimize the disadvantages of body powered prostheses by adding electrically powered support for motions deemed difficult to be carried out by body powered units, with EMG being the controlling interface [14, 17]. As it shown in the diagram below, the hybrid active prosthesis consist of both electronic controlling equipment and body powered prosthetic hook as the terminal device for better grasping and manoeuvring of objects. The electronic controlling mechanism controls the cables connecting body powered prosthetic hook with upper arm. Hook has considered as more reliable grasping mechanism because the existing prostheses fingers are still under development stage. Subject has the ability to operate two different joints such as elbow and shoulder at once with cable, but limited to once motion at a time for body powered prostheses [14, 16].

Externally powered prostheses are more advanced and they are controlled using elec-



Figure 2.5. Active hybrid prosthesis arm with shoulder joint

tronic muscle signals generated by remaining muscles of the amputee. These signals are known as electromyographic (EMG) signals. Power source is generally a lightweight power bank and the controlling is done by the EMG signals. Due to the advancement of technology and the initial cost, externally powered prostheses are considerable more expensive to develop. Control sequence consist of all the possible motions for each EMG signal and researchers need to control all the motions and limitations. Inability to follow the path during extension after flexion as in body powered or hybrid prostheses is one of the disadvantages in externally powered prostheses. For such motion the extension motion has to be separately actuated by the motors [14, 17]. Motions such as flexion angle of upper arm with shoulder also has to be controlled by the motor controller using algorithms [16, 18].



Figure 2.6. Active externally powered prosthetic arm with shoulder joint

2.2.5 Shoulder Socket

Shoulder socket connects the artificial prosthetic hand to the shoulder. Most existing shoulder sockets do not provide adequate comfort to the subject for prolonged periods of use. The shoulder socket shown in the figure below connects the prosthesis barely by a single strap which transfers the force of weight of prosthesis and any other load held by the prosthesis, and this thin strap can easily result in discomfort [14]. Pain incurred by shoulder sockets will be an adverse effect for long term prosthesis use, and is commonly stated as a deterrent to the use of active prostheses by users in surveys. If the strip runs over any sensitive area of the body such as a rib, then the subject will suffer from immense pain. Apart from comfort, socket has to be stable and strong enough to hold the complete load. Notable problems with shoulder sockets are weight of the socket, heat buildup within the socket, lack of stability, and, improper contact with prosthesis and the shoulder socket. Shoulder sockets can result in discomfort to the user when prosthesis is used for prolonged periods, hence the aim should be upon minimizing the discomfort when using the prosthesis [14, 16].



Figure 2.7. Body powered shoulder socket, which can result in discomfort to user

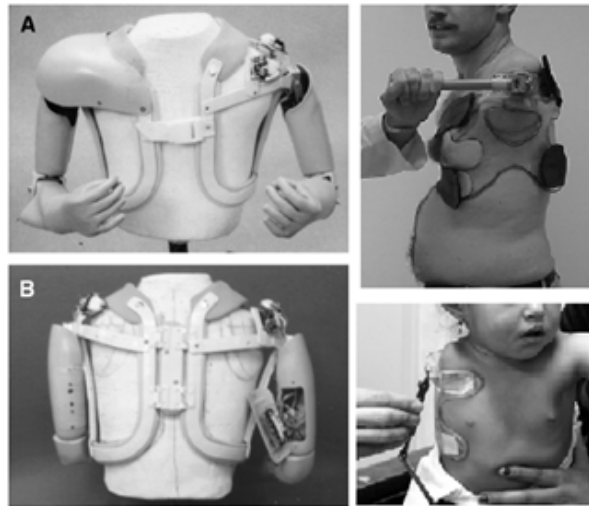


Figure 2.8. Newly developed shoulder sockets with emphasis on comfort

Multiple design criteria need to be assessed before design of shoulder socket. The socket design should prioritize the need of amputee to wear the prosthesis for long periods without discomfort. Most initially developed sockets cover the entire girdle and most of the trunk. Those were bulky and resulted in heat buildup due to lack of ventilation. Later designs focus more on ergonomics, such as the designs shown in the figure below. The fabric or plastic material used have sufficient porosity for ventilation. The symmetric shape results in the load being evenly distributed all over the contact area. Additional fabric padding is provided for sensitive areas, providing better dampening of contact between socket and body. Socket with locking mechanism is another modern design which has attached to a child shown in below figure. The shoulder socket is an important component in the development of a shoulder prosthesis, as it acts as the connection between the body of the user and the prosthesis, and enables the functioning of the device [14, 15, 17, 19].

2.2.6 Degree of Freedom

Ability to Major parameters of the prosthetic device during motion are calculated using the data obtained from the degree of freedom angles. These parameters are position analysis, velocity analysis and acceleration analysis. Velocity and acceleration analyses are derived from position analysis matrix [16, 20]. Three basic shoulder joint motions have to be replicated in a shoulder joint prosthesis. They are flexion-extension, abduction-adduction and internal rotation-external rotation. During each of these motions, torque is generated inside the muscles, which creates a resultant force. When developing artificial prosthesis researchers use this resultant force signal and amplify the signal in order to move the artificial prosthesis [16, 20].

2.2.7 Actuators

Actuation is governed by motors. Due to the weight of prostheses, as well as the torque needed to control the motion of prostheses, motors with a high torque / weight ratio should be used. Degree of freedom of shoulder is generally considered as three. If all three degrees are to be implemented as active degrees of freedom, three motors are usually used in the shoulder joint of prosthetic device for individual directional movement. Direct current is utilized as the energy source for the motors, since the motors utilized are powered by direct current. Out of the motor types available - brushed / brushless DC motors, stepper motors and servo motors - DC motors are more suitable for this application in order to obtain continuous torque during the movement [16, 18, 21].

The major disadvantage of brushed DC (BDC) motors is the size of the motor increasing significantly with the increase of torque from motor, because of the increment of number of coils. When the motor becomes larger, the weight of the prosthetic device increases, and this is not favourable. Comparatively lightweight brushless DC (BLDC) motors are more commonly used in prostheses applications. Most of the existing shoulder prostheses have used flat BLDC motors because of this reason. Flat motors are smaller by size, allowing them to be fitted in the prostheses. Gear arrangement of the motor increase the torque. Generally flat BLDC motor torque values are below milli-newton meter (mNm) region. In order to increase torque, gears have been used in existing prostheses [22, 23]. Separate BLDC motor controllers are used to control each motor. Motor controllers receive the signal from microcontroller. The control hardware contains the function algorithm. Inputs are received by the controller hardware from EMG signal acquisition unit after signal conditioning is carried out [16, 18, 21, 23–26].

2.2.8 Control Hardware

In externally powered type prostheses, most of the prostheses motions are controlled by BLDC motors and control hardware's functionality is control the motor controller by providing electrical pulse as programmed. Raspberry pi, arduino, beagle board are some of the commonly used products. Most projects have used Arduino controller, possibly owing to the open source platform of the controller which result in the widespread availability of knowledge required for Arduino based control (especially on the worldwide web). Ability to use already developed programs is another reason for popularity of Arduino [16, 25–29].

2.2.9 Safety Features

Externally powered prosthetic devices are generally powered by motors. Hence, safety precautions must be introduced to control motors. Safety considerations are integral in most designs. Limit switches or current sensing switches need to be used to shut off the current if the joint exceeds the specified degree of motion. Mechanical stoppers can be introduced as well, as a backup safety precaution. Manual safety mechanisms should also be present for the safety of the user in events such as sudden electrical failures. Frictional brakes are also used by some products such as Otto bock arm as a safety measure. However there are some improvements to be made in those mechanisms such as controllability and secondary safeguard mechanism [16, 20, 30].

2.3 Control of Active Shoulder Prosthetics

2.3.1 Introduction

Main challenge with today's prostheses is their low controllability and low functionality. The ideal prosthesis would be easy to control and comfortable to wear. Still most of the commercially available prosthetic devices haven't been able to generate the natural movements of a hand. A better control method to achieve a good controllability has become a must needed requirement.

Multiple approaches are used for the control of externally powered prosthetic devices. The output from the external drivetrain which powers the prosthesis can be controlled by the user, by using analogue or digital sensors.

Electromyography signals are among the most useful electro-physiological signals for both engineering and medical applications. However, it is challenging to overcome disturbances caused by various types of noises recorded from raw EMG signals. In order to achieve a more standardized and precise signal analysing and classification of signal is very important. In this literature review an EMG based control method for active shoulder prosthetics will be discussed.

2.3.2 High Level Control

A control signal is critical for the appropriate control of a prosthetic device. An active prosthesis is generally powered by a motor. Thus the control signal in consideration should be able to control the output of an electric motor, a DC motor in practically every situation. This control

signal can be a digital signal – which can turn the motor on and off, and / or an analogue signal which can vary the speed and direction of the motor. Different control methodologies have been adapted in acquiring this control signal, from the basic on – off switch to body signal based control (EMG and EEG).

2.3.2.1 Types of Body Signals Used

Electromyography (EMG) and Electroencephalogram (EEG) are two high level prostheses control methods which are utilised in powered prostheses. EEG is defined as an electrical activity generated by brain that recorded from the scalp surface [31]. EMG (Electromyography) is the process of detection, analysis and utilisation of electrical signals generated by skeletal muscles [32]. The electrical signals thus generated are known as myoelectric signals which represents the anatomical and physiological properties of muscles. EMG signals are being used for a vast range of applications in applied research, sports biomechanics, biomechanical research, rehabilitation [33].

Prosthetic devices

Electromyography signals are detected with the help of EMG electrodes. Mainly there are two types of EMG electrodes:

- Surface electrodes
- Inserted electrodes

Inserted electrodes can be divided into two more types: needle and fine wire electrodes. Wire electrodes are less painful when the electrodes are inserted and detect electric signals from many directions, but needle electrodes pick up electric signals individually. These methods need strict medical supervision and certification [32].

Surface electrodes access EMG signals from the surface above the muscle, which makes their use simple and painless. Due to the simplicity of this method surface EMG is highly applicable to the prostheses requirements of physically disabled people [34].

Whenever an EMG signal is being recorded from the muscle, various background noises are received due to the presence of electronic equipment and physiological factors. Therefore, analysing and classifying the EMG signals is considerably difficult due to the complexity of EMG signals.

Inability to remove the noise completely from the signal remains an unsolved problem. Therefore, that EMG signal is processed and analysed to get the required information to a

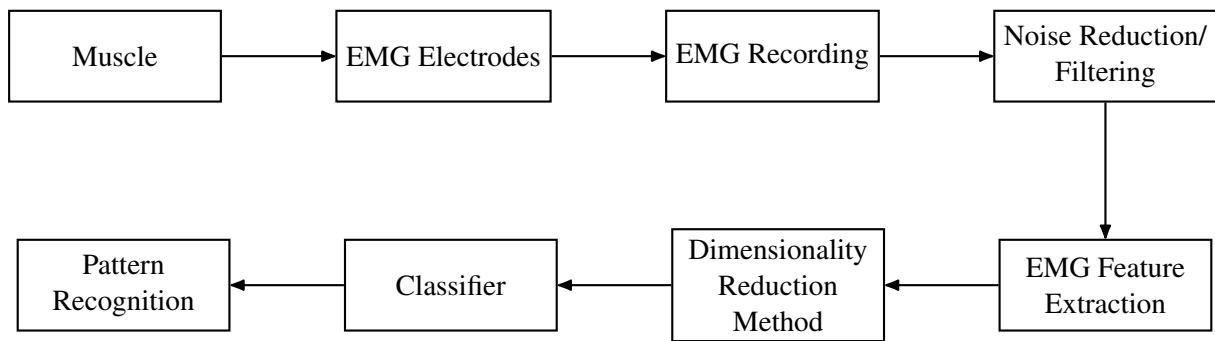


Figure 2.9. EMG Signal Classification Process

satisfactory level. Researchers have used multiple methodologies, procedures for analysing the EMG signals appropriately [33].

2.3.3 EMG signal acquisition

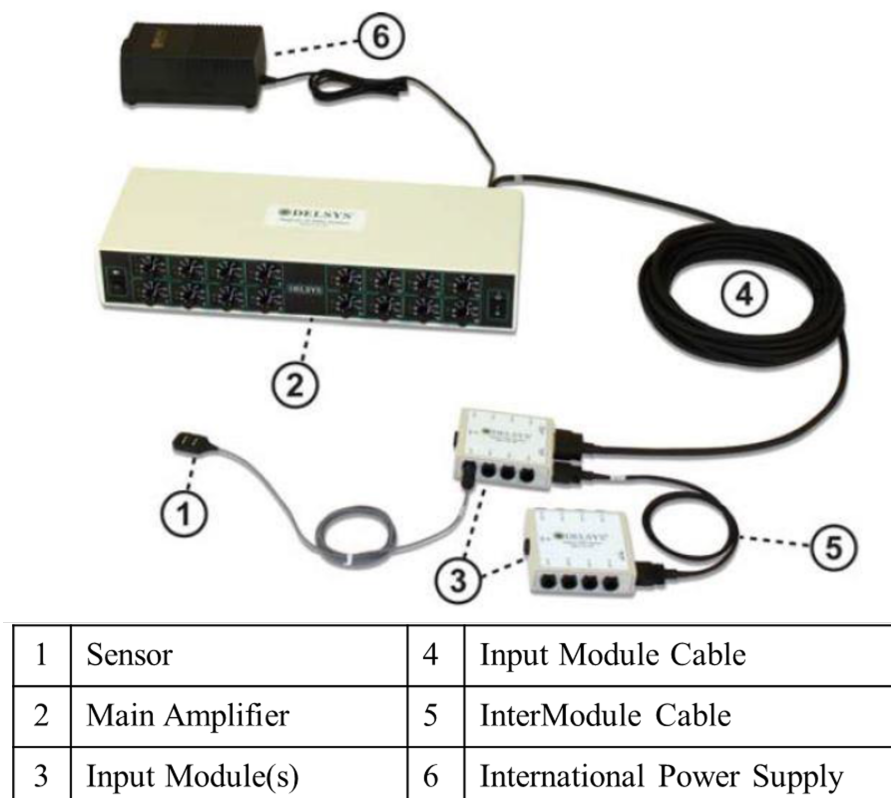


Figure 2.10. The components of the surface EMG signal acquisition system

Special skin preparation is required before the application of surface EMG electrodes in order to obtain a stable electrode contact and low skin impedance (5 k Ω to 50 k Ω). Removal of the hair and dead cells on the skin is necessary to improve the adhesion of the electrodes. For gelled electrodes, an Ag-AgCl layer is applied, but for dry electrodes gel is not required [34].

Preferred electrode location for signal acquisition is in the midline of the belly of the muscle between the nearest innervation zone and the myotendonous junction. The EMG signal with the greatest amplitude is detected at this zone [35]. EMG signal is obtained through differential amplification technique. This is achieved with the help of an instrumentation amplifier with a high input impedance. Some amplifier configuration methods are:

- Monopolar Configuration
- Bipolar Configuration
- Multipolar Configuration [32]

Monopolar configuration is a simple method carried out using only a single electrode, but this method is not a recommended method for EMG signal acquisition owing to its high level of crosstalk. Bipolar configuration is the most commonly used electrode configuration. This method is a more accurate method compared to monopolar configuration. Two electrodes are used for signal acquisition with a reference electrode and they should be placed at a distance of 1 to 2 cm from each other. Multipolar configuration uses more than two surface electrodes. This method is a more accurate method which further reduces the crosstalk and noise [32].

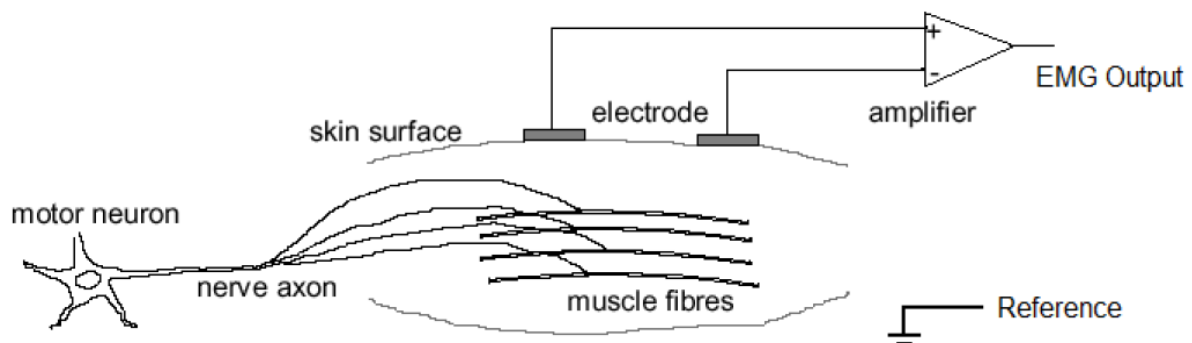


Figure 2.11. Bipolar Configuration

2.3.4 Signal Filtering and Amplification

Raw EMG signal is weak and interfered with electrode noise, electrode and cable motion artefact, alternating power line interference, and broadband noise from electronic instrument [36]. Researchers have used different types of processing techniques for cancelling these noises. Proper use of these techniques can increase EMG signal quality where the signal becomes much more accurate, simple, and reliable [32].

Filtering allows only a specific range of frequencies. The frequency range which is filtered out is called the stop band and the range which is allowed is called the passband. Filters

can be categorized as low pass, high pass, band pass and band stop according to their behaviour [37].

Low pass filter eliminates the high frequency component from the EMG signal which is higher than the cut-off frequency (f_c) [38]. Low pass filtered signal is detected across a capacitor. High pass filter removes the low frequency component from the EMG signal. The high pass filtered signal is gathered across a resistor. After the acquisition the signal is filtered generally using a band-pass filter with high CMRR (Common Mode Rejection Ratio) and gain. In order to reduce motion artefacts, a high-pass filter and to cut off the noise a low-pass filter is used. In general, approximately 95% of the power range of the EMG signal is distributed up to 400 Hz, and most of the remaining part is electrode and equipment noise. A low-pass filter is usually applied to the signal. The common choice for cut-off frequency is 500 Hz. A high-pass filter is also required to cut off movement artefacts and the instability of the electrode–skin interface with a cut-off frequency of approximately 10 Hz [34, 39].

Filtered signal has to be amplified to a suitable level. This can be easily achieved by using a non-inverting amplifier. A weak EMG signal of approximately 1-10 mV amplitude should be amplified with a minimum gain of 500-1000. A higher gain, in the order of 10000, is required for the muscles with a weak EMG response [32].

2.3.5 Feature Extraction

Since raw EMG signals are not suitable as input signals to the controller, the feature of the raw signal must be extracted [40]. The fundamental purpose of feature extraction is to establish the important information in the measured signal while rejecting noise and irrelevant data. Different EMG feature extraction methods like time domain, frequency domain, and time-frequency domain are used by researchers. Time domain includes features such as mean absolute value, mean absolute value slope, variance of EMG, frequency ratio which can be calculated quickly. Most of these features have been widely used in research [31]. Time frequency representation allows a more accurate way to localize the energy of the signal in both time and frequency. TFR requires more computationally heavy transformation. Among different TFR methods short-time Fourier transform (STFT), wavelet transform (WT) and wavelet packet transform (WPT) are highly preferable [31, 39].

2.3.5.1 Fourier Transform

Fourier transform is a frequency analysis method. The inability of the Fourier transform to describe both time and frequency characteristics of the waveform led to a number of different approaches [41].

2.3.5.2 Short Time Fourier Transform (STFT)

The concept of STFT is to divide the waveform into a number of short segments and performing the analysis on each of these segments, usually using the standard Fourier transform. This method has been successfully applied in a number of biomedical applications [41].

2.3.5.3 Wavelet Transform

Wavelet transform overcomes the main drawback of the STFT by varying the time–frequency aspect ratio and by producing a good frequency resolution. This method is widely used as an efficient tool to extract useful information from the EMG signal. It is a time-frequency analysis method which provides more information than the original signal [42]. Wavelet transform method is divided into two types as discrete wavelets transform (DWT) and continuous Wavelet transform (CWT) [43, 44].

2.3.6 Signal Classification

Performing the motion intention of the user precisely, based on EMG signals in real time is not a simple task to achieve. Due to the following practical obstacles it (EMG signal) requires a good signal classification technique.

- EMG signals differ each time for the same motion even with the same person since the EMG signal is a biologically generated signal.
- The activity level of each muscle and the way of using each muscle for a certain motion differ from subject to subject.
- Real-time motion prediction is not very easy since many muscles are involved in a joint motion.
- One muscle is not only concerned with one motion but also with other kinds of motion.
- The activity level of some muscles are affected by the motion of the other joints [40].

Among several possible EMG signal classification techniques, most used are Artificial Neural Networks (ANN) and Fuzzy classifiers [45].

2.3.6.1 Artificial Neural Networks

An artificial neural network (ANN) is a computational system inspired by the learning characteristics and structure of biological neural networks. ANN should be trained with a large amount of data taken from a subject. ANNs possess several attractive features that make them suitable for difficult signal processing problems [39].

- Ability to learn from experience without requiring a mathematical model of the underlying signal characteristics
- Ability to adapt to changing environmental conditions
- Ability to process degraded and incomplete data [39]

2.3.6.2 Fuzzy Logic

Fuzzy logic has the ability to deal with imprecise, uncertain and imperfect information. The strength of fuzzy logic is its ability to develop human-made systems that can perform information processing in a manner similar to the brain [39, 46]. Fuzzy logic systems can match human decision-making more closely than ANNs, because of the possibility of introducing the knowledge of an expert in the fuzzy rules in the form IF-THEN [39].

2.3.6.3 Neuro-Fuzzy Systems

Fuzzy logic and neural systems have very different application requirements. Fuzzy systems are more appropriate when sufficient expert knowledge about the process is available. Neural networks are more useful when sufficient data are available or measurable. Both these approaches build nonlinear systems, either numerical or symbolic. However, fuzzy systems show both symbolic and numeric features. Therefore, the integration of neural and fuzzy systems leads to a “symbiotic” relationship in which fuzzy systems provide a powerful framework for expert knowledge representation, while neural networks provide learning capabilities and exceptional suitability for computationally efficient hardware implementations. Neuro-fuzzy computing enables us to build more intelligent decision-making systems by combining the advantages of ANNs (massive parallelism, robustness and learning in a data-rich environment) with the fuzzy modelling of imprecise and qualitative knowledge.

2.3.7 Summary of Literature Review

In this section, an attempt was made to highlight the important facts about the shoulder anatomy, amputations in general and upper limb amputations (including statistical data) and prosthetic solutions available for shoulder amputees. Out of upper limb amputation types, it is understood that most amputees utilize passive prostheses owing to the advantages of passive prostheses (lightweight, cosmetically superior and overall ease of use), despite the lack of functionality. Hence the active prosthesis solution that is intended to be developed should address the inherent shortcomings of active prostheses available for amputees at present.

Externally powered prostheses are more advanced prostheses and they are still at research level and not affordable for most people due to the high cost. Shoulder joint prostheses are three degree of freedom prostheses. In a natural shoulder these motions are mainly governed by a ball joint, accompanied by the motion of acromioclavicular joint. However no method has been developed for accurate control of a ball joint. Hence the three different directions of motion have to be taken into consideration separately and be controlled by using separate motors. Safety precautions must be in place to prevent any damage to the user in case of malfunction of prosthetic device.

By analysing the work carried out in development of externally powered prosthetic solutions for amputees, it can be concluded that active shoulder socket has to be designed with priority given to the comfort of the amputee. Flat brushless DC motors are more suitable as they comparatively light weight and high torque. Based on the available resources for the project group, Arduino can be considered as an appropriate microcontroller based controlling platform for the controlling of the prosthetic device, as a wide variety of open source resources is available.

Many researches have been carried out on prostheses control methodologies. As prostheses interact with human users, the control method used should be reliable, accurate and safe. When these factors are taken into consideration, surface EMG can be identified as the most suitable method for signal acquisition from amputees for control of prostheses. Unlike intramuscular EMG acquisition, this method is not painful for the user and can be used repeatedly.

Despite being a good method to control prosthesis, surface EMG signal acquisition has its own limitations which needs to be minimized or eradicated for proper utilization. A proper signal processing method should be used in order to overcome these issues. Several undesired signal sources such as extrinsic factors, inherent noise in electronic equipment, motion artefacts, and ambient noise can be minimized to a great extent with signal classification.

As discussed in this literature review few steps have to be followed to process the signals. Extracted signals should be filtered and suitable feature extraction method is applied. After that signals are classified using a suitable classification technique.

However, there are few unavoidable limitations in EMG signals because it is a biological signal. Even for the same movement signal characteristics can be different according to the subject; it can even vary on the physical condition of the subject. Therefore a good preparation and practice is required to overcome these issues.

Chapter 3 - Hardware Construction of Modular Artificial Shoulder Prosthetic Device

The research and development work related to this project were carried out in order to achieve the goals explicitly stated in the objectives – To develop a 3 DOF shoulder prosthetic device, and control. From the literature review, it was observed that most standalone shoulder prosthetic devices developed up to now were either under-actuated, or featured abductor and flexor axes with considerable lateral distance. The aim was to overcome these issues, and develop a novel shoulder prosthesis. Furthermore, this is the first time the design and development of a shoulder prosthesis had been undertaken by the Bionics laboratory of Department of Mechanical Engineering, University of Moratuwa.

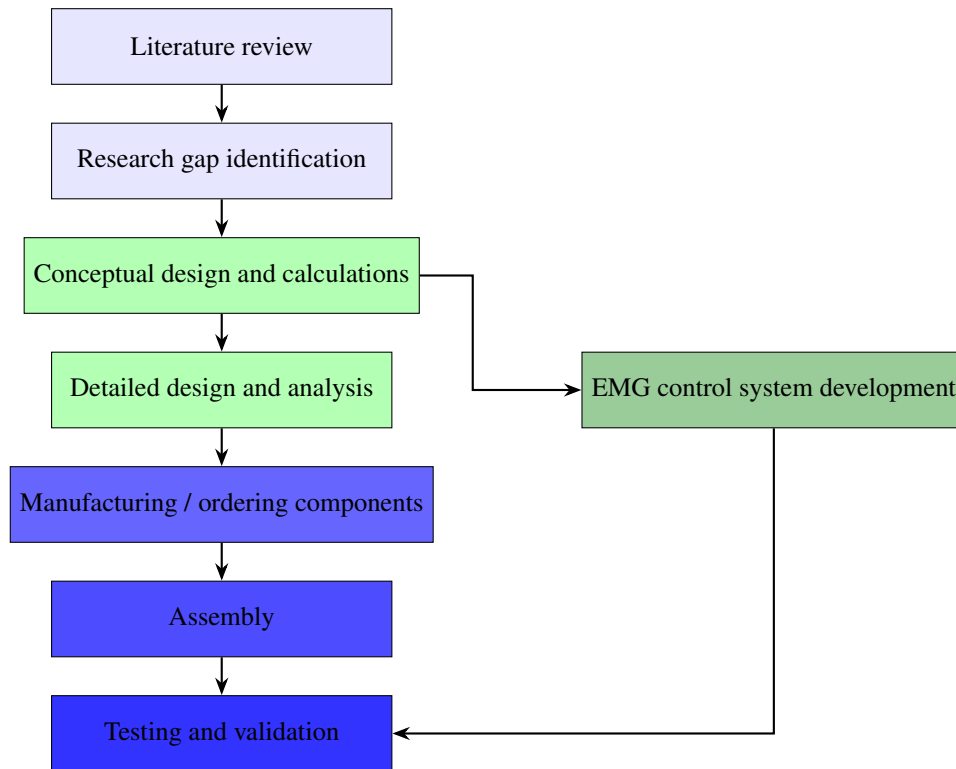


Figure 3.1. Process Diagram for the Implementation of the Shoulder Prosthetic Device

3.1 Design Requirements and Considerations

The main design requirement for the shoulder prosthetic device was to have three active degrees of freedom to replicate the motion of a human shoulder. The motion limitations are as follows;

Table 3.1. Design requirements for the shoulder prosthetic device

	Human shoulder	Proposed shoulder prosthesis
Degrees of Freedom	3	3
Motion ranges (degrees)		
Flexion - Extension (a)	180:(-30)	180:(-30)
Abduction - Adduction (b)	150 : 0	150 : 0
Internal-External rotation (c)	+90:(-90)	+90:(-90)
Average torques (Nm)		
Flexion - Extension (a)	18	24
Abduction - Adduction (b)	18	24
Internal-External rotation (c)	3.5	6
Average velocity	N/A	5 rpm = 30° / second

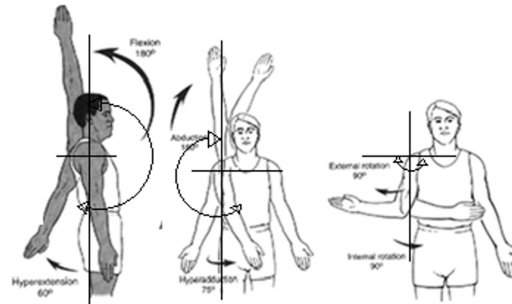


Figure 3.2. Motions of the Shoulder Complex – (a) Flexion – (b) extension, abduction – adduction, and (c) internal rotation – external rotation

In order to achieve this motion range, a prosthetic device with an electric motor as the prime mover for each axis, was proposed.

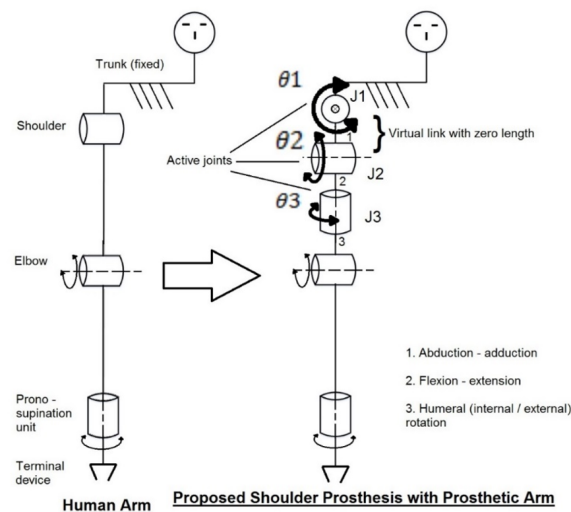


Figure 3.3. Kinematics diagram of the proposed shoulder prosthesis

The design was now focused on implementing a method to attach motors within the prosthetic device to allow the required motions. In addition, the following design constraints were identified;

1. The axes for abduction and flexion motions should preferably have their axes intersecting each other, or at least a minimum lateral distance between them, to allow for the best simulation of the functioning of the actual human shoulder. The axis for humeral rotation need not have such a design constraint.
2. All three axes should be active, which meant that three different motors had to be used, with an appropriate torque multiplication mechanism.

3.2 Motor Selection

One of the critical hardware elements in the shoulder prostheses is the motor. The motor acts as the device which provides motion for the degrees of freedom of prostheses. The prostheses needs a total of three motors for three different degrees of freedom. The following criteria are considered in selecting a suitable motor

- Dimensions and weight of the motor
- Maximum torque provided by the motor
- Cost of the motor
- Usable lifespan of the motor/reliability
- Power requirements of the motor

The preliminary analysis showed that brushless type DC motors were the most suitable motor type for the application. After consideration of the above criteria, the available options were studied, in terms of manufacturers and types. Two major manufacturers of brushless DC motors which are frequently utilized in biomechanical applications were noted

- Maxon
- Faul – Haber

3.2.1 Motor Selection Calculation

Minimum torque requirement for flexion and abduction was calculated based on an average person (50th percentile American male of age 40 years) [45].

$$\text{Body weight of the person} = 70 \text{ kg}$$

$$\text{Height of the selected person} = 1.8 \text{ m}$$

$$\text{Additional weight in the hand (for ADL)} = 0.5 \text{ kg}$$

$$A\text{Gravitational acceleration} = 9.8 \text{ m s}^{-2}$$

$$\text{Safety factor} = 1.2$$

For both abduction and flexion motions, the maximum torque is generated when the hand is in fully extended configuration (perpendicular to the trunk)

According to the studies of Bernstein et al [40]

$$\text{Weight of hand as a percentage of total body weight} = 1.41 \%$$

$$\text{Weight of upper arm as a percentage of total body weight} = 5.31 \%$$

$$\text{Weight of forearm as a percentage of total body weight} = 3.64 \%$$

$$\text{Weight of hand as a percentage of total body weight} = 1.41 \%$$

According to the studies conducted by Harless [40]

$$\text{Length of upper arm as a percentage of total body height} = 21.11 \%$$

$$\text{Length of forearm as a percentage of total body height} = 17.1 \%$$

$$\text{Length of hand as a percentage of total body height} = 11.76 \%$$

According to the studies of Bernstein [40]

Distances to the centre of mass from the proximal joint (as a fraction of total length)

$$\text{Upper arm (shoulder)} = 0.466 \%$$

$$\text{Forearm (elbow)} = 0.412 \%$$

$$\text{Hand (wrist)} = 0.506 \%$$

Table 3.2. Torque Calculation for the Abduction / Flexion Motions

Arm Segment	Length %	Length (m)	CM %	CM (m)	Total distance to CM from shoulder (m)	Weight (%)	Weight (kg)	Force (N)	Torque (Nm)
Upper arm	21.11	0.37998	0.466	0.177071	0.17707068	5.31	3.717	36.7277	6.503395
Forearm	17.10	0.30780	0.412	0.126814	0.30388428	3.64	2.548	25.1768	7.650830
Hand	11.76	0.21168	0.506	0.107110	0.41099436	1.41	0.987	9.75255	4.008242

Net torque generated = Σ (Torques of the upper arm, forearm, and hand)

$$= 18.16 \text{ Nm}$$

With safety factor of 1.2 = 21.79 Nm

$\therefore$ Maximum torque requirement with the safety factor = 21.79 Nm

Out of the two initially selected manufacturers, there were two motors which were dimensionally acceptable for the prosthetic device

- Maxon EC 45 Flat brushless motor
- Faulhaber Series 2610 brushless flat micromotor

Based on the availability and cost, it was decided to use Maxon EC 45 flat brushless DC motor, which is a 50W motor with diameter 42.8 mm and nominal maximum continuous torque of 83.4 mNm. The crucial reason for choosing this particular motor was the OEM gearhead available for the motor. When the motor was coupled to the Maxon GS 45 gearhead (89:1 reduction), the combined maximum output torque is 2 Nm.

Since this combination alone is inadequate to satisfy the torque requirement, an additional gearbox was required. Harmonic Drive gearboxes are generally used for power transmission and torque amplification in prosthetic devices, however due to financial constraints, it was not possible to obtain a Harmonic Drive unit. Therefore it was required to design a gear drive to get the required torque.

Table 3.3. Motor specifications (Maxon EC 45 Flat 4 pole brushless DC motor)*¹

Nominal voltage	24 V
No load speed	6710 rpm
No load current	185 mA
Nominal speed	5240 rpm
Nominal (max. continuous) torque	83.4 mNm
Nominal (max. continuous) current	2.33 A
Stall torque Stall current	780 mNm
Stall current	23.3 A
Max. efficiency	83.3 %

Gearhead (Maxon GS 45 Spur Gearhead)*¹

- Max. continuous torque – 2 N m
- Max. intermittent torque – 2.5 N m
- Max. continuous transmittable power – 11 W
- Efficiency – 66 %

*¹ - All figures are sourced from the manufacturer



Figure 3.4. Maxon EC 45 4 – pole flat BLDC motor with Maxon GS45 gearhead

3.3 3D Model Design

All 3D models were created using Solidworks design software. 3D CAD models were developed after initial rough sketching work was carried out.

3.3.1 3D Model Design Phase I

The first conceptual model focused on the motors which were selected, and a simple method of connection to provide the required motion. However, the design was deemed to be imprac-

tical with gears which could be very much larger than what could be accumulated within the proposed shoulder prosthesis, and linkages which were practically difficult to fabricate. Hence this design was substantially changed thereafter.

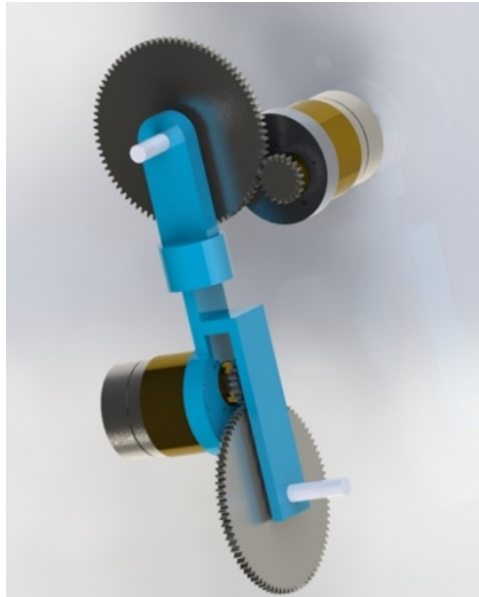


Figure 3.5. 3D model design phase I

3.3.2 3D Model Design Phase II

The second design was a conceptual design which focused on the motion requirements and limitations, and was not a detailed design. This design was carried out to study an arrangement of gearbox and motor to obtain the required motions, after being attached to the shoulder.

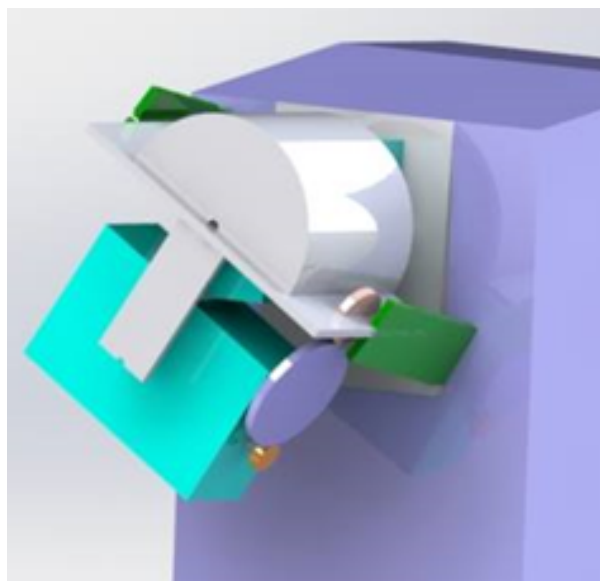


Figure 3.6. 3D model design phase II

3.3.3 3D Model Design Phase III

After further analysis, work was started on the third design iteration. This was intended to be developed over time and eventually become the final design. The design incorporated the following main features;

- 3 spur gearboxes, 2 with a reduction of 1:12 (abductor and flexor units), which result in an overall torque of 24 Nm, and 1 with a reduction of 1:3 (humeral rotator), resulting in a torque of 6 Nm.
- Structural members which permit the 3DOF motion and other major design requirements.
- The gearboxes for abduction and flexion motions which would permit the achievement of design constraint 1.

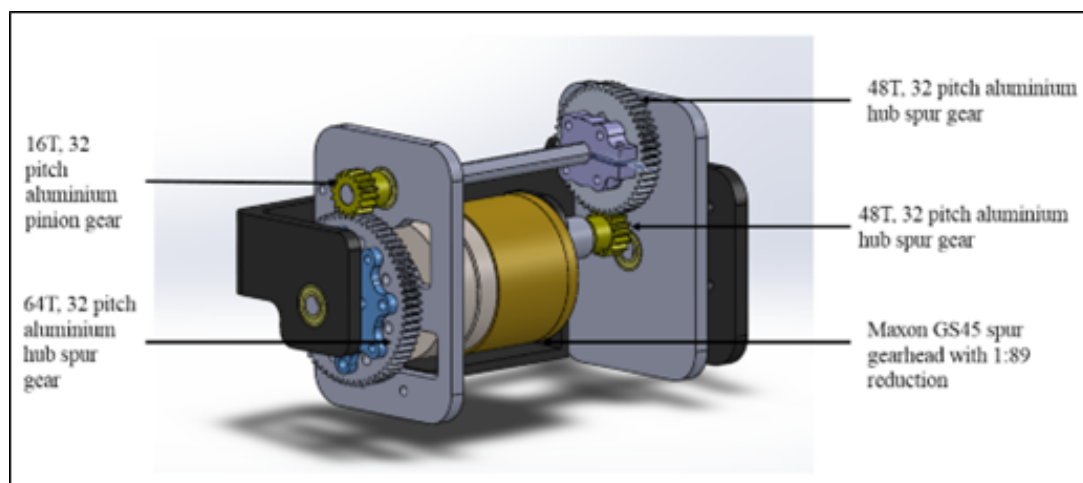


Figure 3.7. The gearbox used for abductor/ flexor units, with an overall 1:1068 reduction



Figure 3.8. Isometric views of the final 3D CAD design

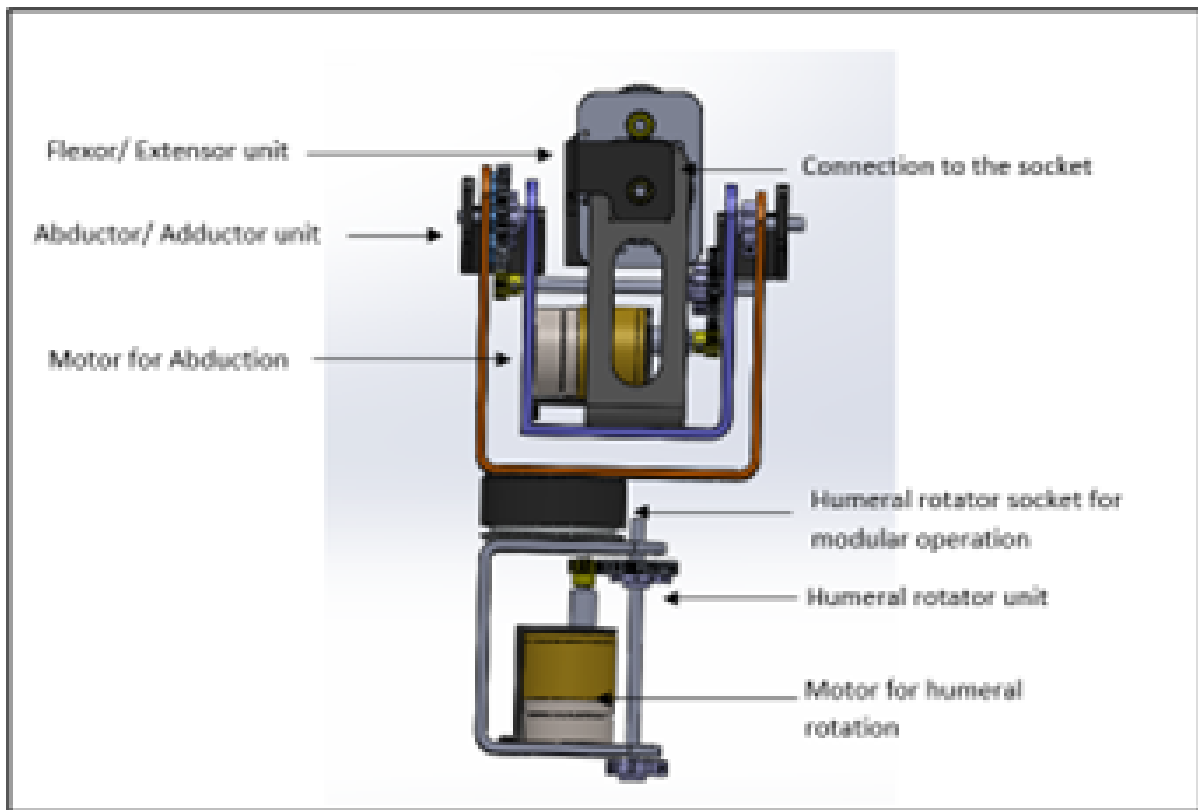


Figure 3.9. Components of the prosthetic device

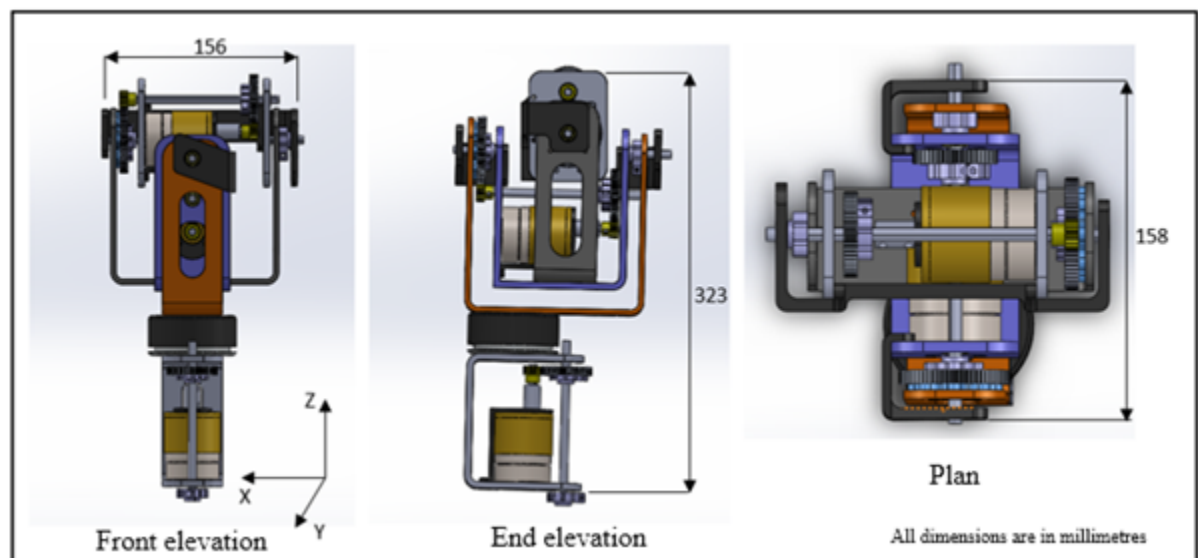


Figure 3.10. Dimensions of the prosthetic device

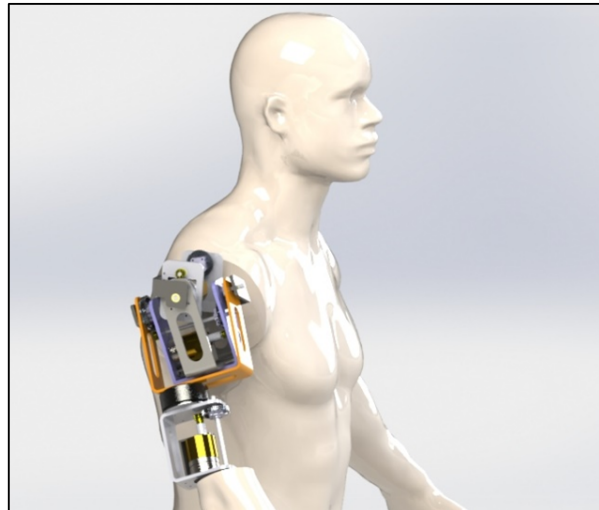


Figure 3.11. Isometric view of CAD model, with human figure for comparison

3.3.4 Mechanical limits

Certain mechanical limits were integrated to the design to limit the motion of the prosthesis, in order to prevent unnecessary, unrealistic motion for a shoulder complex.

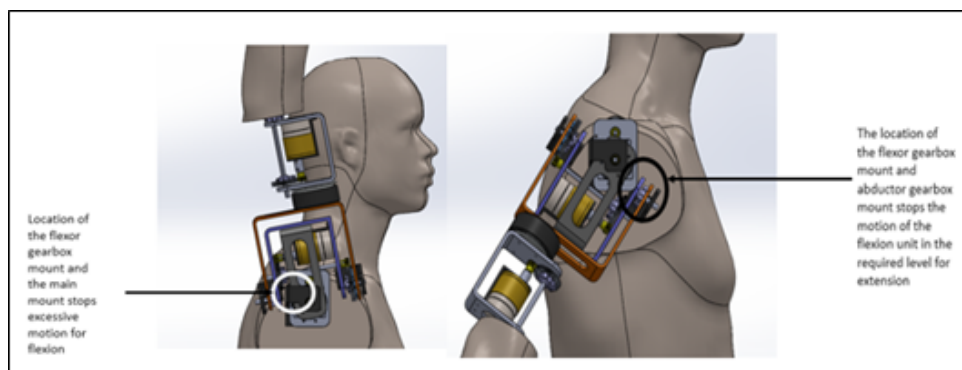


Figure 3.12. Mechanical limits for flexion – extension motions

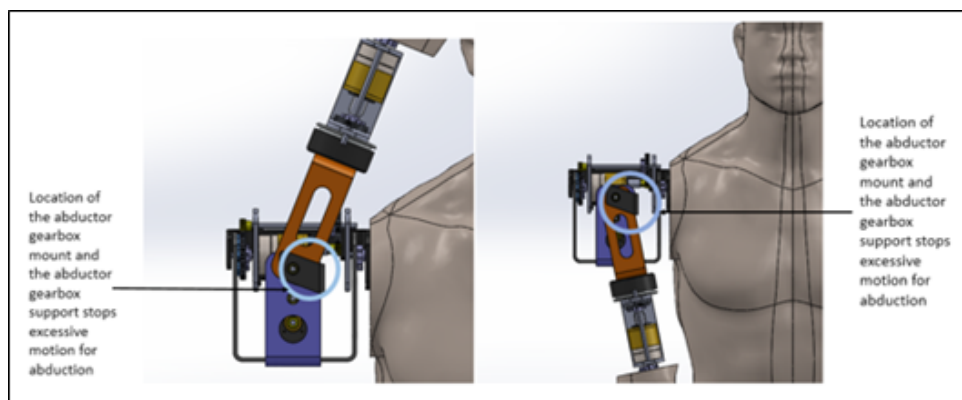


Figure 3.13. Mechanical limits for abduction – adduction motions

3.3.5 Perceived advantages and disadvantages of the design

Advantages

- Activation of all three degrees of freedom with maximum torque requirement satisfied
- The positioning of the axes for abductor and flexor motions with a minimal lateral distance

Disadvantage

- The width of the prosthesis (from the front elevation) is 162 mm. This could result in a pronounced protrusion from the suspension point, more than that of a natural human shoulder (which does not protrude beyond the humerus). This can present an unnatural image when used in an actual situation.

3.4 Components

When the design was finalized, it was decided to manufacture the structural components and source the drivetrain components from a specialist supplier, in order to prevent any problems associated with drivetrain misfits and inaccuracies during manufacturing.

3.4.1 Fabrication

3.4.1.1 Material Selection

The following material options were available for the construction of structural components:

- Stainless steel

Although stainless steel offered acceptable strength for this purpose, the available stainless steel types were denser compared to available aluminium. Since weight is a critical factor for the comfort of the user, stainless steel was not selected.

- Carbon Fibre Reinforced Plastic (CFRP)

Multiple types of CFRP such as PAN and rayon are available for the fabrication of individual components. The notable advantage of CFRP is its excellent strength to weight ratio next to metals and alloys. However the cost of CFRP, and the lack of a flexible fabrication location as well as the difficulty in reworking and post fabrication adjustments were the reasons to not select CFRP.

- 3D printed plastics

The strength of plastics used for 3D printing, poly lactic acid (PLA) or acrylonitrile butadiene styrene (ABS) were deemed inadequate for the structural components.

For the structural members, the primary material chosen was aluminium, owing to its advantageous features such as low density, high strength, ease of machining and acceptable cost. Furthermore, the details of currently developed prostheses had used aluminium (type unspecified), or CFRP for structural components.

3.4.1.2 Manufacturing

The following structural components were required to be manufactured

- Motor mount (3 units)
- Mount 1 – Mount 6
- Bearing support (2 units)
- Gearbox mount 1 - 2
- Socket for humeral rotator

The components were manufactured from aluminium sheets (type 6061) of thickness 3 mm and 6 mm. Most components were manufactured using CNC machinery (in the Die and Mould Facilitation Centre, Department of Mechanical Engineering). The reason for choosing CNC machining over conventional machining process was primarily the level of accuracy required from the components. With most parts requiring proper alignment (as gearbox supports) and minimum gap between moving components, the tolerance was minimal. Hence CNC machining was selected as the machining method.

For components where CNC machining was deemed not feasible (3 components which could not be manufactured by single aluminium sheets and bent as the other components), owing to their structural importance. The component “mount 1” is the crucial load bearing structure in the prosthetic robot. Hence, the previous manufacturing method of CNC machining and bending could have reduced the strength of the component (due to the strength reduction caused by bending).

Furthermore it was not possible to machine them with a single aluminium block as that would result in an impractically high level of material wastage. Hence for these three components, sand casting with aluminium was selected as the manufacturing method.

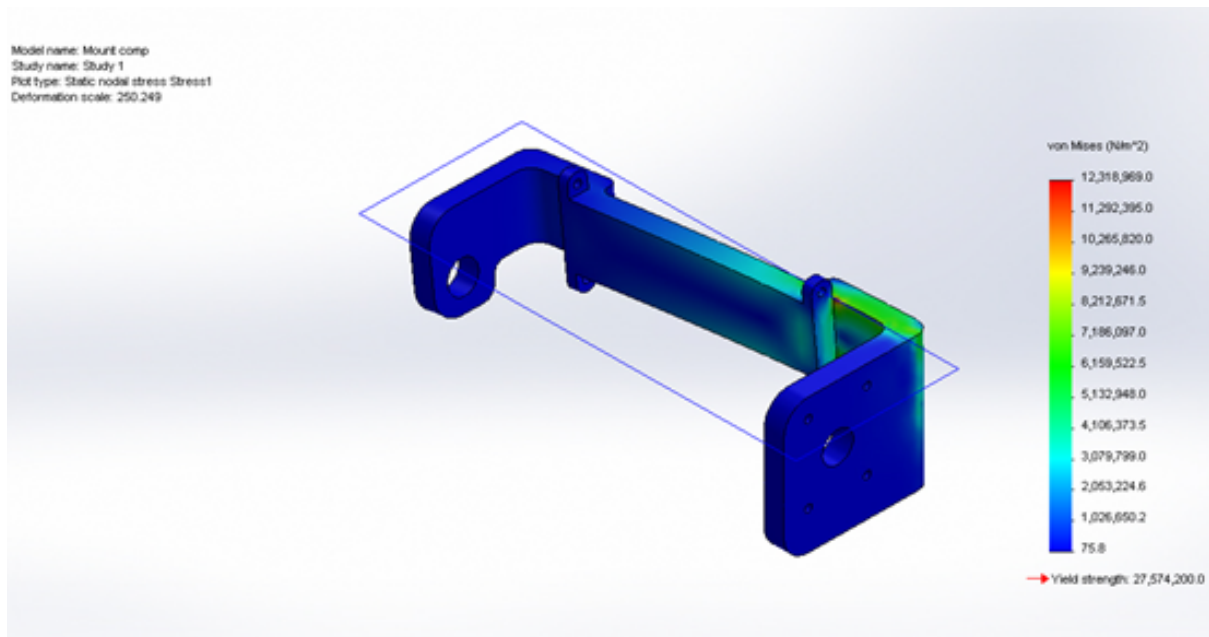


Figure 3.14. FEM analysis for the load bearing component of the prosthesis (mount 1)



Figure 3.15. Patterns for the casting of mount 1 and other cast component

After casting was completed, machining was carried out to improve the surface conditions.

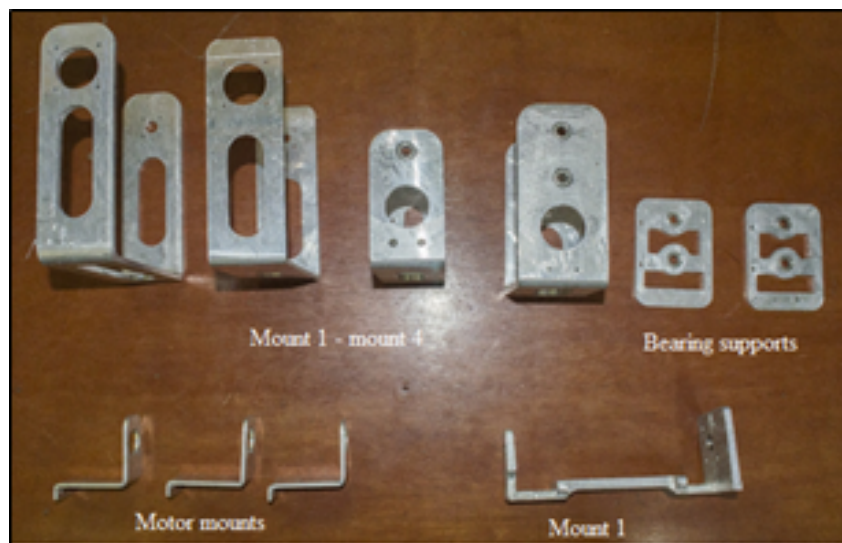


Figure 3.16. Manufactured components

3.4.1.3 Purchased Components

The following components were sourced from Servo City (United States of America).

- 16T pinion gear
- 64T, 32 pitch aluminium hub spur gear
- 48T, 32 pitch aluminium hub spur gear
- 1/4" bore clamping hub
- 1/4" to 8mm set screw shaft coupler
- 1.50" x 0.770" Hub Adaptor
- 1" shaft
- 1.25" shaft
- 1.5" shaft
- 5" shaft
- Non-Flanged Ball Bearings .25" ID, .5" OD

The spur gears had been manufactured with aluminium type 7075 which is aeronautical grade aluminium, widely used for aerospace industry. The shafts and bearings are made of stainless steel.

In order to facilitate the finished prosthesis to be erected, a separate support structure is to be assembled using automation aluminium profiles.

3.5 Assembly

The components were then assembled, and mounted on a frame specifically designed to support the prosthetic device. For the frame, aluminium extrusions used for automation projects, were used, to allow for convenient positional adjustment. Furthermore the frame was designed with a considerable weight at the bottom to counter the torque generated by the prosthesis.

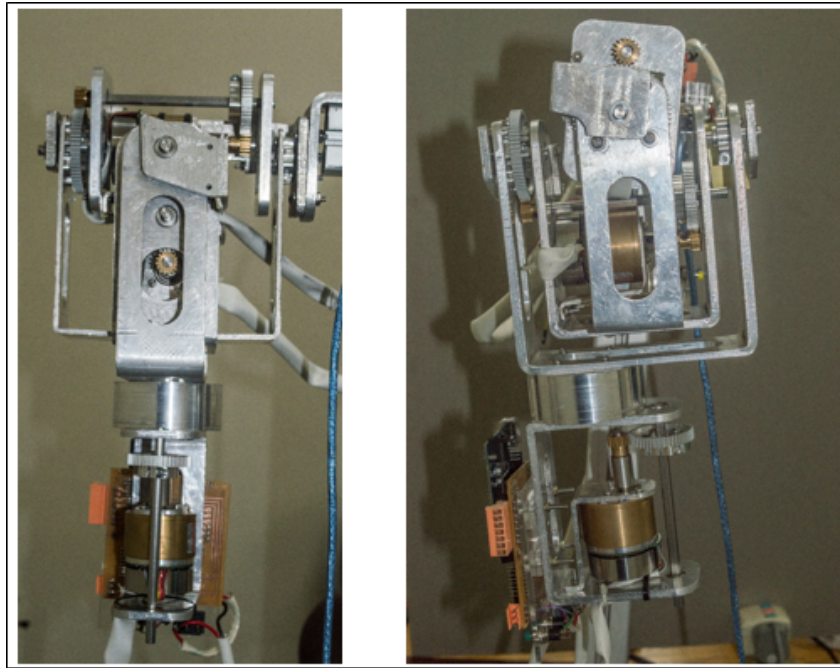


Figure 3.17. Front elevation and side elevation of the assembled prosthesis



Figure 3.18. The assembled prosthesis on the aluminium support structure

The completed prosthesis (with motors and gearbox) weighs 3.2 kg.

3.6 Modular Functionality

The prosthetic device was intended to be designed as a modular prosthetic device. This should allow the humeral rotator to function as a separate unit, which can be separately mounted and controlled independent of the abductor and flexor unit. The reason to allow independent operation of humeral rotator is to cater the needs of trans - humeral amputees who possess a short residual humeral stump which is not capable of providing torque for humeral rotation, but can provide abduction – adduction and flexion – extension.

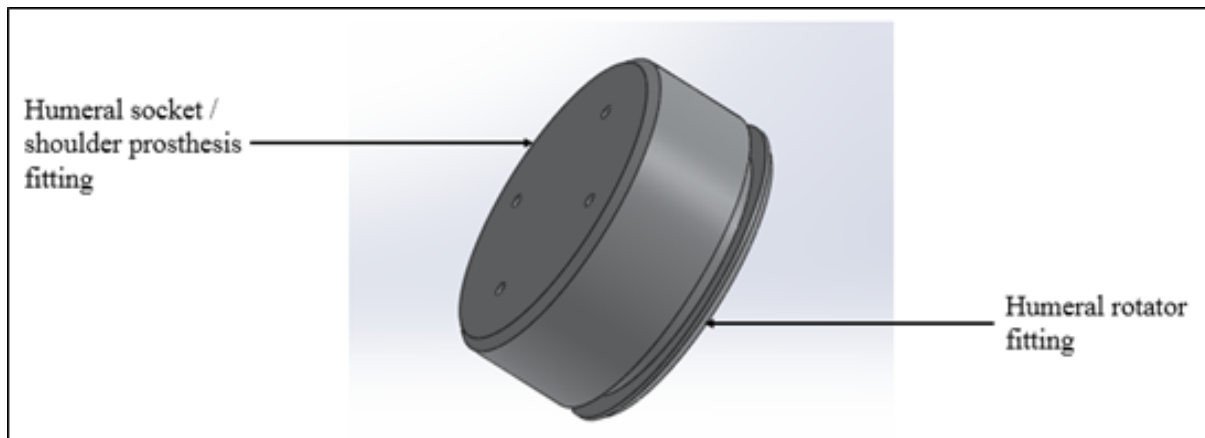


Figure 3.19. Socket for the humeral rotator, allowing for modular feature of the prosthesis

The socket was designed as a threaded joint to allow convenient attachment and removal of the humeral rotator as required.

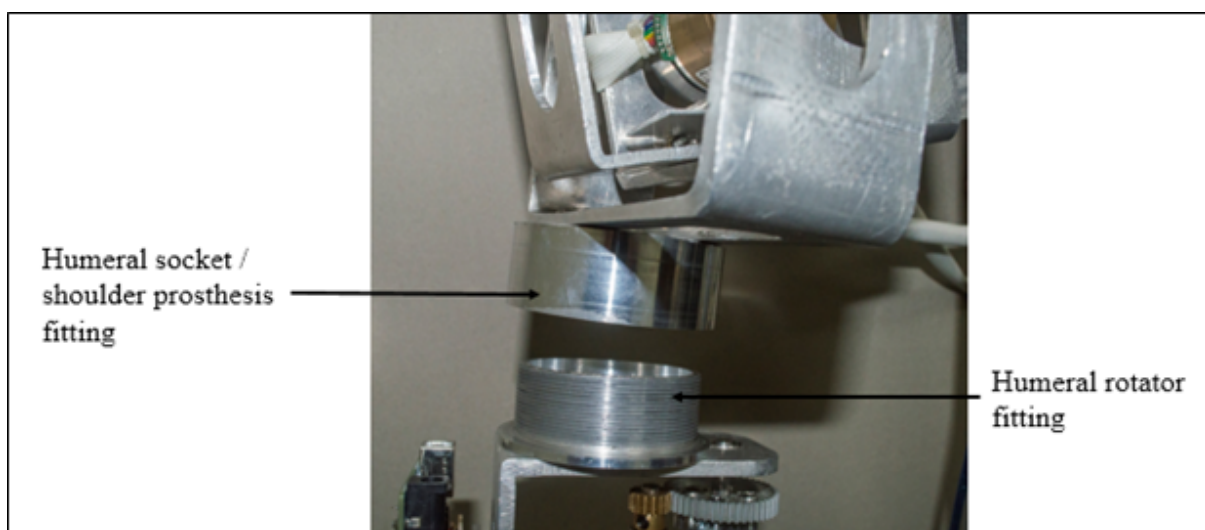


Figure 3.20. Fabricated socket attached to the prosthesis

3.7 Design of Shoulder Socket

Another important part of implementing a prosthesis is to develop an interface between the user and the prosthesis. In the case of the shoulder prosthesis, the interface is a shoulder socket, which is mounted on the user and is used to hold and support the prosthesis. The socket needed to achieve the following requirements

- Provide a comfortable interface for the user, to allow extensive use of the prosthesis without discomfort owing to the socket
- Provide adequate support to the prosthesis without undergoing physical change, to facilitate accurate operation and alignment of the prosthesis
- Should facilitate the EMG extraction process, allowing the location of EMG acquisition electrodes and not causing any movement

Furthermore, the material available for the design of socket were also studied. Thus, it was deemed not possible to use a carbon fibre socket design. The structural components were intended to be fabricated by either 3D printing polymer or by using aluminium. With the above considerations, the design of the shoulder socket was carried out.

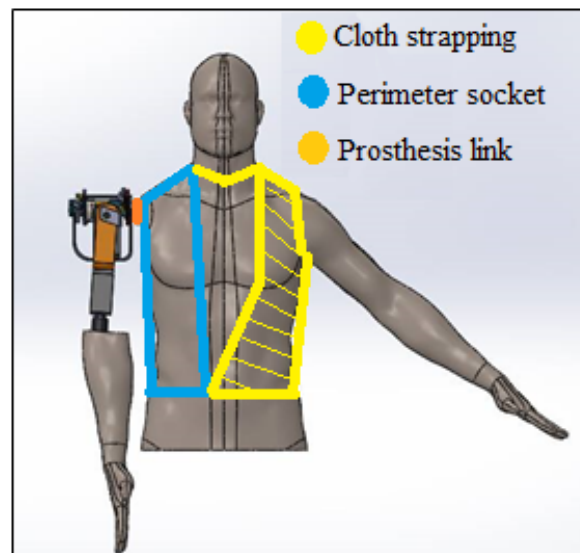


Figure 3.21. The conceptual design for shoulder socket: The conceptual design was carried out with the available material in mind (aluminium)

3.7.1 Conceptual design

In the first iteration, a simple design to support the prosthesis was developed. A ‘perimeter frame’ that could be fabricated with aluminium was designed. In order to ensure user comfort, a fabric lining would run right beneath the perimeter.

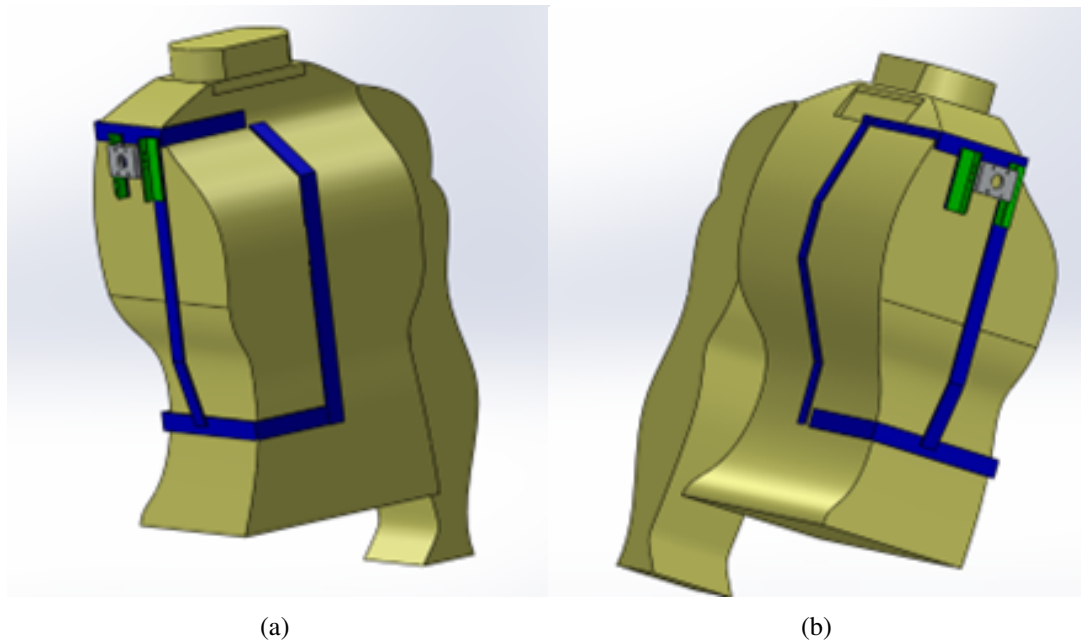


Figure 3.22. Isometric views of the conceptual 3D design

In the updated design, the structural components except for the load bearing prosthetic mount, were designed to be manufactured using 3D printing.

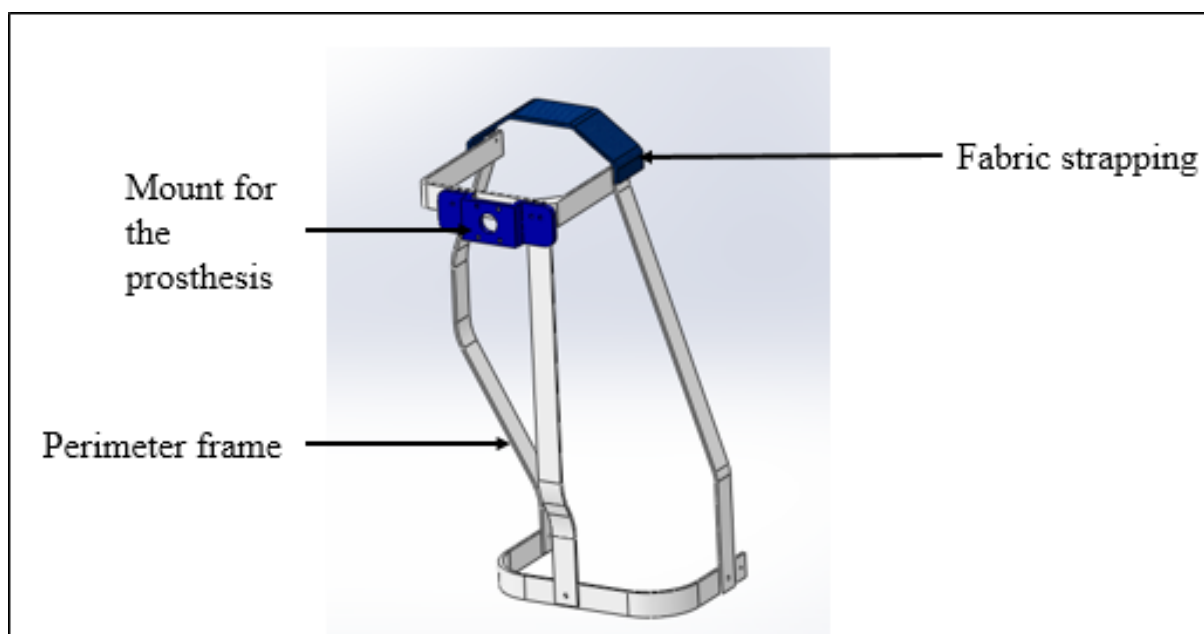


Figure 3.23. 3D design for the perimeter frame type shoulder socket

Chapter 4 - Control of the Modular Shoulder Prosthetic Device

4.1 Control hardware

4.1.1 Motor Controllers

In order to control the selected motor, OEM motor drivers were purchased to reduce the problems associated with the low reliability and incompatibility of non – OEM motor controllers.

There are several OEM motor controlling units available for the control of this motor.

- Maxon EPOS2 50/5 digital positioning controller (unit price CHF 599.70 $\approx$ Rs. 86356/-)

The EPOS2 50/5, according to the manufacturer, is a modular digital positioning controller, which can be used with brushed DC motors with encoder, and also brushless EC motors with Hall sensors and encoder, from 5 to 250 watts

- Maxon DEC module 50/5 1-Q-EC amplifier (unit price CHF 74.30 $\approx$ Rs. 10699/-)

The DEC Module 50/5 (Digital EC Controller) is a 1-quadrant amplifier. This controller is intended to be used on brushless motors with maximum voltage of 250 W, and is capable of speed control and PWM operation.

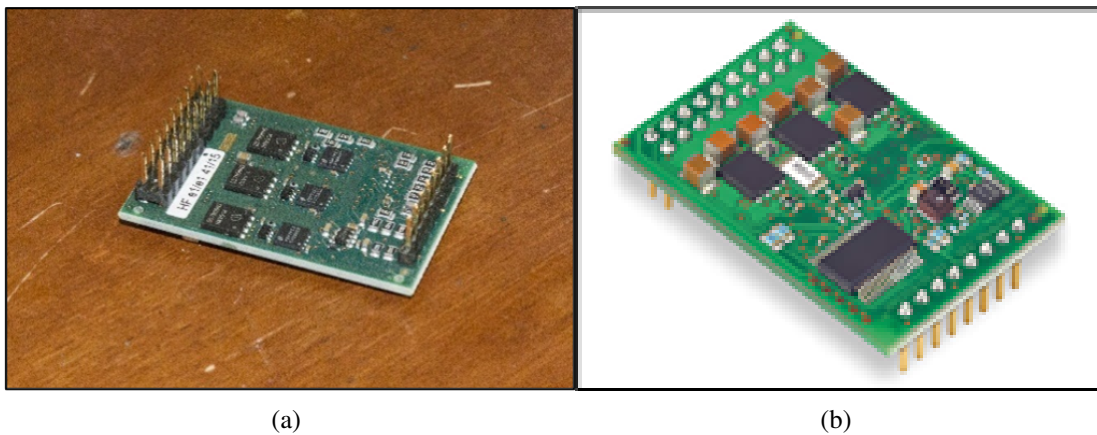


Figure 4.1. Maxon DEC 50/5 module

Out of the two most suited controllers, the DEC module was selected since it was capable of providing the required motion control, even though the motor controlling algorithm building would be much more difficult compared to the EPOS2 controller unit, which allows for much simpler algorithm building. And the critical factor was the price. Since 3 motors were required

Table 4.1. Specifications of the Maxon DEC 50/5 1-Q-EC motor control unit

Nominal output voltage	6 V - 55 V
Continuous output current	5 A
Switching frequency	46.8 kHz
Input set value speeds	Analogue input of 0 – 5 with 1024 steps
No. of Hall sensors as input	3
Output signals	Digital outputs for motor speed and status

to be controlled, 3 controllers were needed. Purchasing 3 EPOS2 controllers was well beyond the financial capabilities.

For the control of the motor, the Arduino microcontroller based development platform was used, to enable input – output handling and coordination. An Arduino Mega 2560 type board was used for the interfacing between the motor controller and the input / control signals as well as the control computer program.

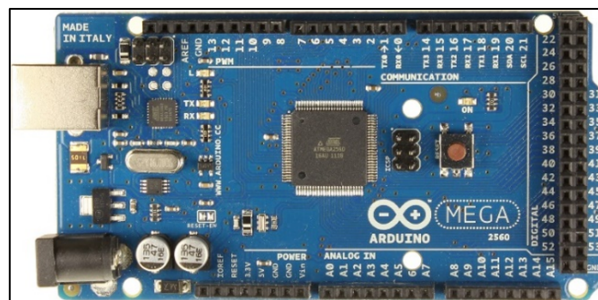


Figure 4.2. Arduino Mega 2560 microcontroller - based development board

The reasons for using Arduino as the microcontroller platform are;

- Convenient to program and implement
- Availability of a wide range of open source support for coding
- Cost is acceptable

A printed circuit board was designed and developed for the circuitry.

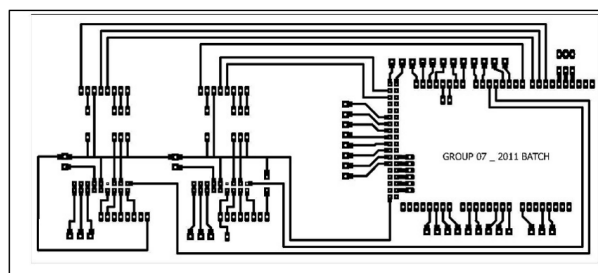


Figure 4.3. PCB design for the shoulder prosthesis circuitry, designed with Proteus software

4.2 Low Level Control

The control system was designed to achieve synchronized motion in all three axes, similar to the motion of the shoulder complex.

The preliminary control was aimed at obtaining controlled motion in the three axes for the abduction – adduction, flexion – extension and internal rotation – external rotation.

The motors used for the prosthesis do not utilize any encoder for position control. In controlling the prosthesis as the requirements, the signal from the Hall sensors of the BLDC motor is utilized by the motor controlling unit. The control of the motor is achieved by using the output from the motor controller unit.

4.2.1 Motor control using DEC controller

The motors used for the prosthesis do not utilize any encoder for position control. In controlling the prosthesis as the requirements, the signal from the Hall sensors of the BLDC motor is utilized by the motor controlling unit. The control of the motor is achieved by using the output from the motor controller unit. The ‘Monitor in’ pin is an output pin which monitors the speed of the motor. The output from this pin is used in the closed loop control of the motor.

Maxon DEC module 50/5 1-Q-EC amplifier consists of 26 pins. Basically control of the motor is handled by 4 pins.

- Monitor in – Monitor the speed and position of motor
- Set value speed (PWM pin) – Speed control of motor
- Direction – Control the direction of motor (clockwise, anticlockwise)
- Enable – Activate the controller

Monitor in pin generates a pulse at each 1/24 segment of the motor. For one revolution monitor pin detects 24 pulses. That means $24 \times 89 = 2136$ pulses for the assembly of motor and the gearbox with a gear reduction of 1:89. Rotation angle and speed of the motor can be monitored by counting the number of pulses detected by monitor in pin.

The ‘Monitor in’ pin is an output pin which monitors the speed of the motor. The output from this pin is used in the closed loop control of the motor.

4.3 High Level Controlling

4.3.1 EMG Acquisition

The shoulder complex functions as a result of the operation of multiple muscles. However, since the surface EMG acquisition is used, only the superficial muscles were accessible for control signal input. Hence, the superficial muscles involved in major shoulder complex motions were identified.

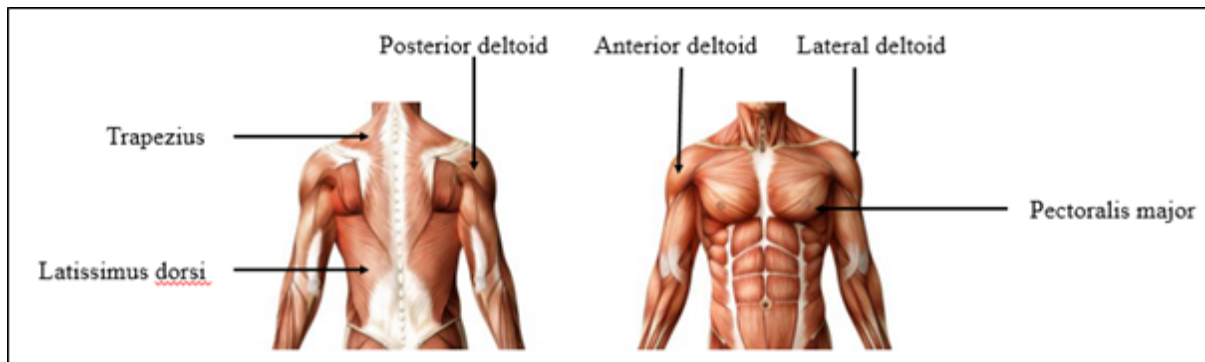


Figure 4.4. Surface muscles involved in the motion of shoulder complex

Table 4.2. Superficial muscles to obtain control signals for shoulder motions

Targeted motion	Prime mover(s)	Other muscles involved
Flexion	Anterior deltoid	Pectoralis major
Extension	Posterior deltoid	Latissimus dorsi, lateral deltoid
Abduction	Lateral deltoid	Trapezius
Adduction	Latissimus dorsi	Pectoralis major, latissimus dorsi
Internal rotation	Anterior deltoid	Latissimus dorsi
External rotation	Posterior deltoid	Infraspinatus

The EMG signals from the deltoid muscles (anterior, lateral and posterior), pectoralis major muscle, trapezius muscle and latissimus dorsi muscle were observed initially, with manual shoulder motion, to identify which muscle signals offer the comparatively best result for the required motion. It was observed that the deltoid muscles displayed pronounced activity for all three types of motions, especially the flexion – extension and internal rotation – external rotation.

Four electrodes were connected to each muscle and reference electrode was connected to stomach area consists lesser amount of moving muscles. Since this data acquisition unit uses the differential signal extraction and amplification electrodes were connected along the muscle.

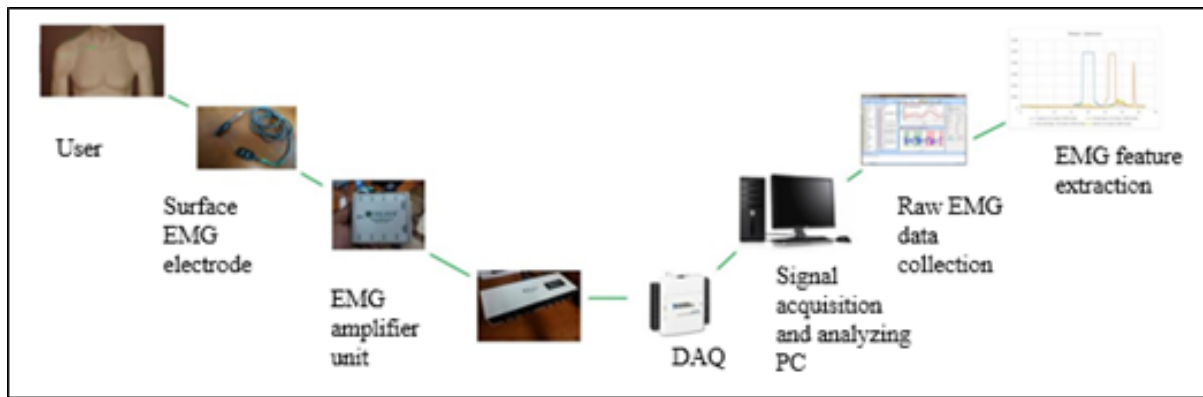


Figure 4.5. Data acquisition process

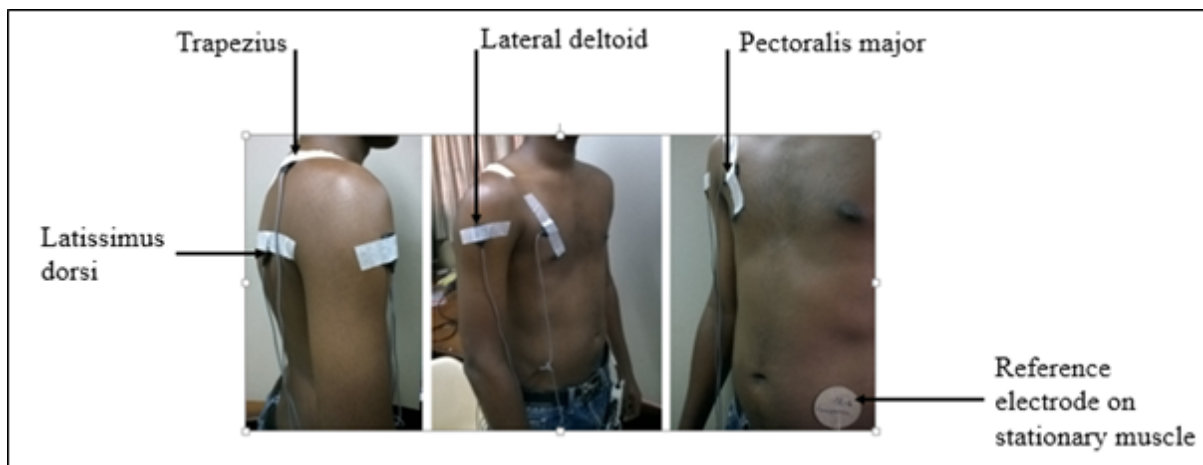


Figure 4.6. Surface EMG electrodes attached to muscles involved in shoulder motion

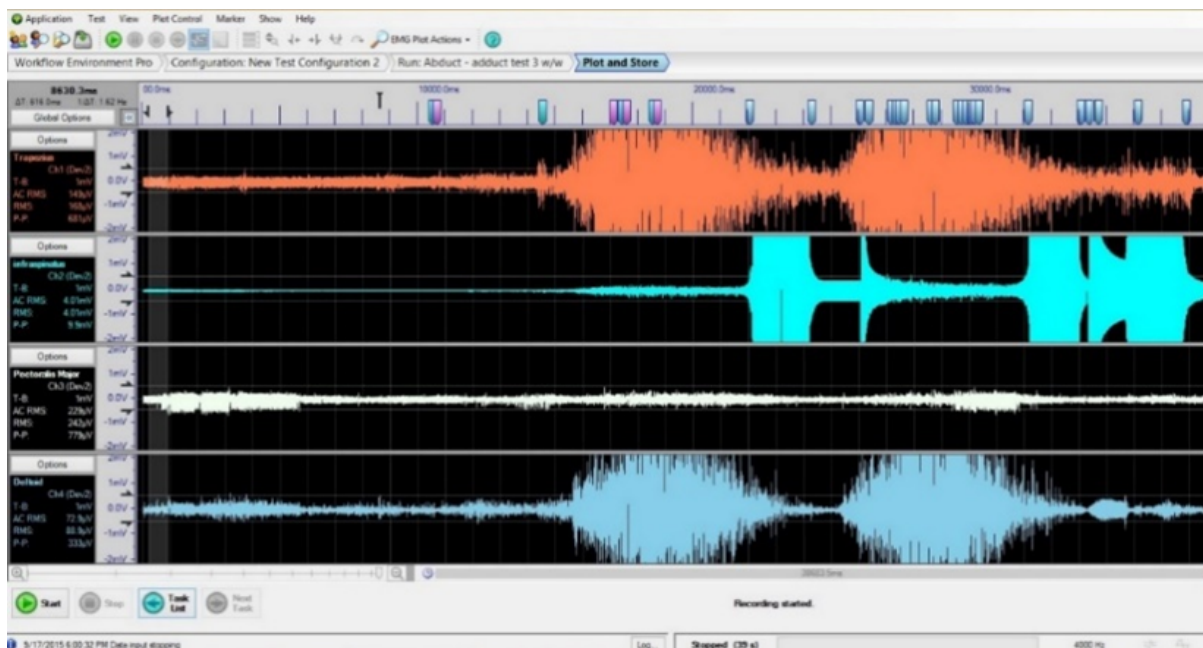


Figure 4.7. Real time raw EMG data extraction using Delsys

EMG signal acquisition was carried out with a healthy 26 year old male. The subject was selected since his EMG signals offered the most clearly observed raw data on the Delsys EMG unit. The sampling rate for data acquisition was set at 2000/s. EMG acquiring channel amplification was set at 1000.

4.3.2 EMG signal processing and control

The raw EMG signals were acquired through the Delsys Bagnoli EMG unit, by using Matlab software, to allow for real time control of the prosthesis using EMG signals.

The raw EMG signals are processed through Matlab, by using Root Mean Square function.

$$\text{Root Mean Square (RMS)} = \sqrt{\frac{\sum_{i=1}^N \text{EMG}_i^2}{N}} \quad (4.1)$$

Where EMG_i is the sampled raw EMG signal voltage at the “ i ” th time interval.

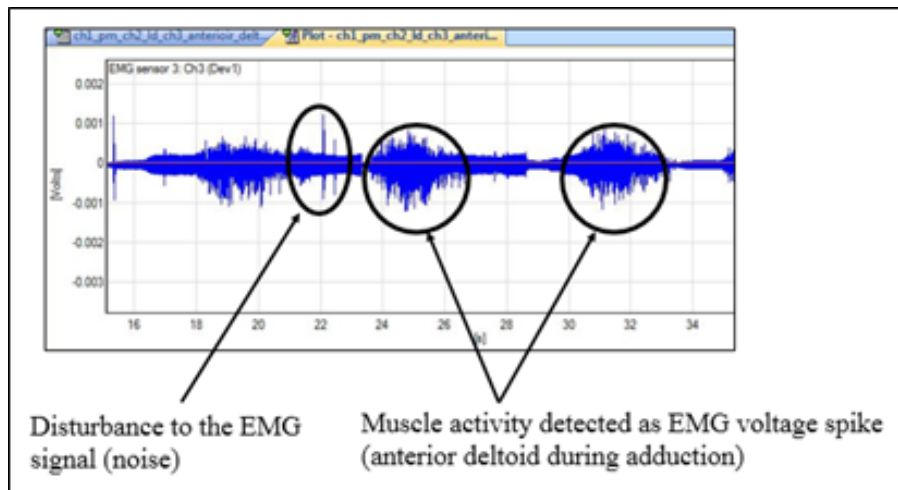


Figure 4.8. Raw EMG signal extracted from the surface EMG electrode

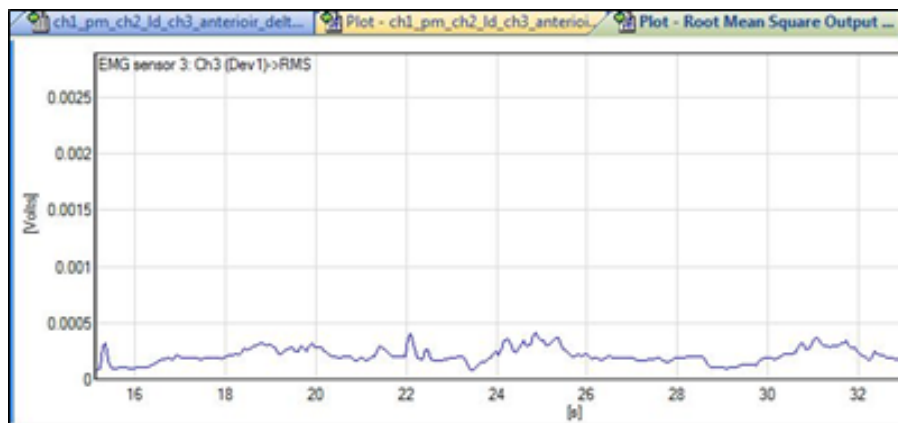


Figure 4.9. RMS value plot of the raw EMG signal

The signal obtained through surface EMG is then converted to PWM values between the range of 0 -255. This PWM value is finally transmitted to the motor control units via serial communication to the Arduino Mega microcontroller board. A PD control method was implemented as follows; $PWM (Motor) = K1.EMG (1) - K2.EMG (2)$ where EMG (1) and EMG (2) are the signals from the agonist and antagonist muscles for the considered motion. As an example, for flexion – extension, the signals from anterior deltoid muscle (agonist for flexion) and posterior deltoid muscle (antagonist for flexion) can be used. K1 and K2 were experimentally found to be 100 for the test subject and the considered muscles (this value can vary alongside the factors that cause variation of the EMG signal). However, the use of EMG signals can cause issues owing to several factors. The voltage of a myoelectric signal is very weak, and depends on a multitude of factors, including the surrounding environmental parameters (temperature, humidity) as well as physical and mental traits of the user. Therefore, a large variation of the signal can occur from person to person. Furthermore, the slightest disturbance to the surface EMG electrode can result in a large variation of the data collected.

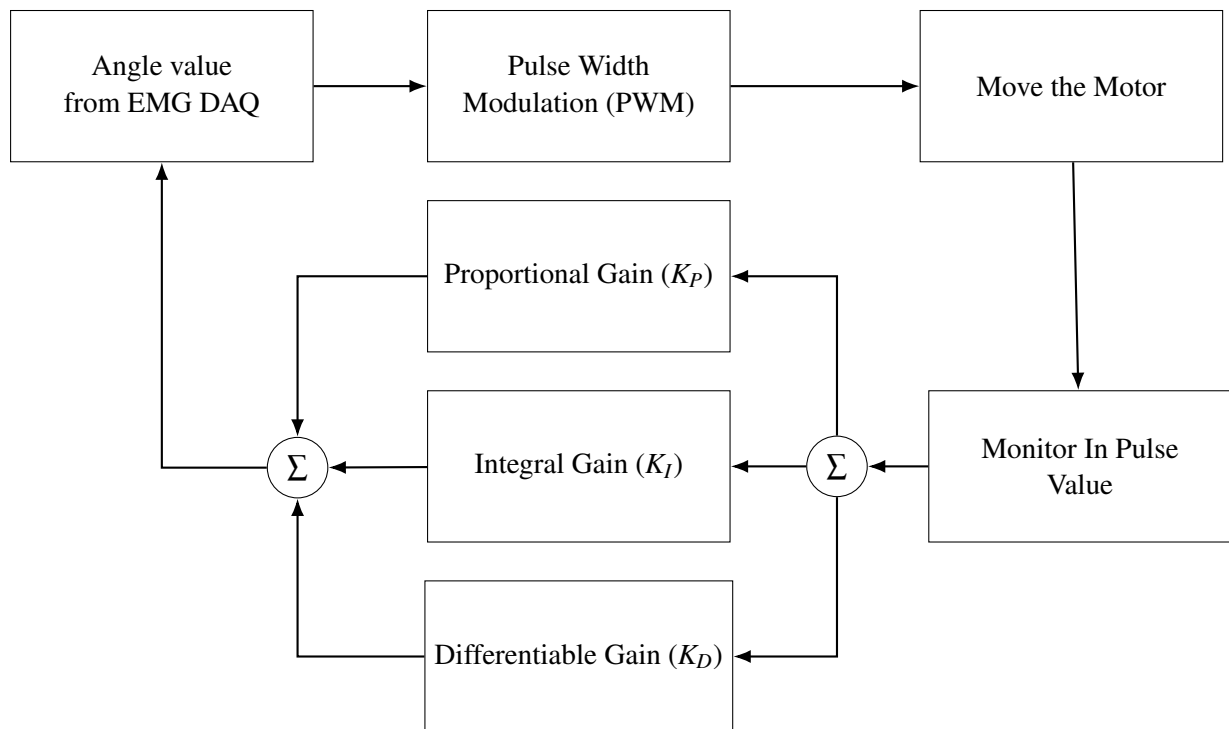


Figure 4.10. Control Loop

Chapter 5 - Experiments and Results

The results of the prosthetic device is analysed based on the motion range of the device, and accuracy of its control system.

5.1 Motion ranges for the actual prosthesis

Table 5.1. The motion ranges of human shoulder, proposed prosthesis and actual prosthesis

Criterion	Human shoulder	Proposed shoulder prosthesis	Actual shoulder prosthesis
Degrees of Freedom	3	3	3
Motion ranges (degrees)			
Flexion - Extension	180:(-30)	180:(-30)	150:(-20)
Abduction - Adduction	150 : 0	150 : 0	208:(-12)
Internal-External rotation	+90:(-90)	+90:(-90)	+90:(-90)
Lateral distance between the abductor and flexor axes (mm)	0	0	14.7

The reasons for the differences in the expected and actual ranges of motion are

- Friction between links and joints
- Dimensional accuracy and tolerances in fabrication

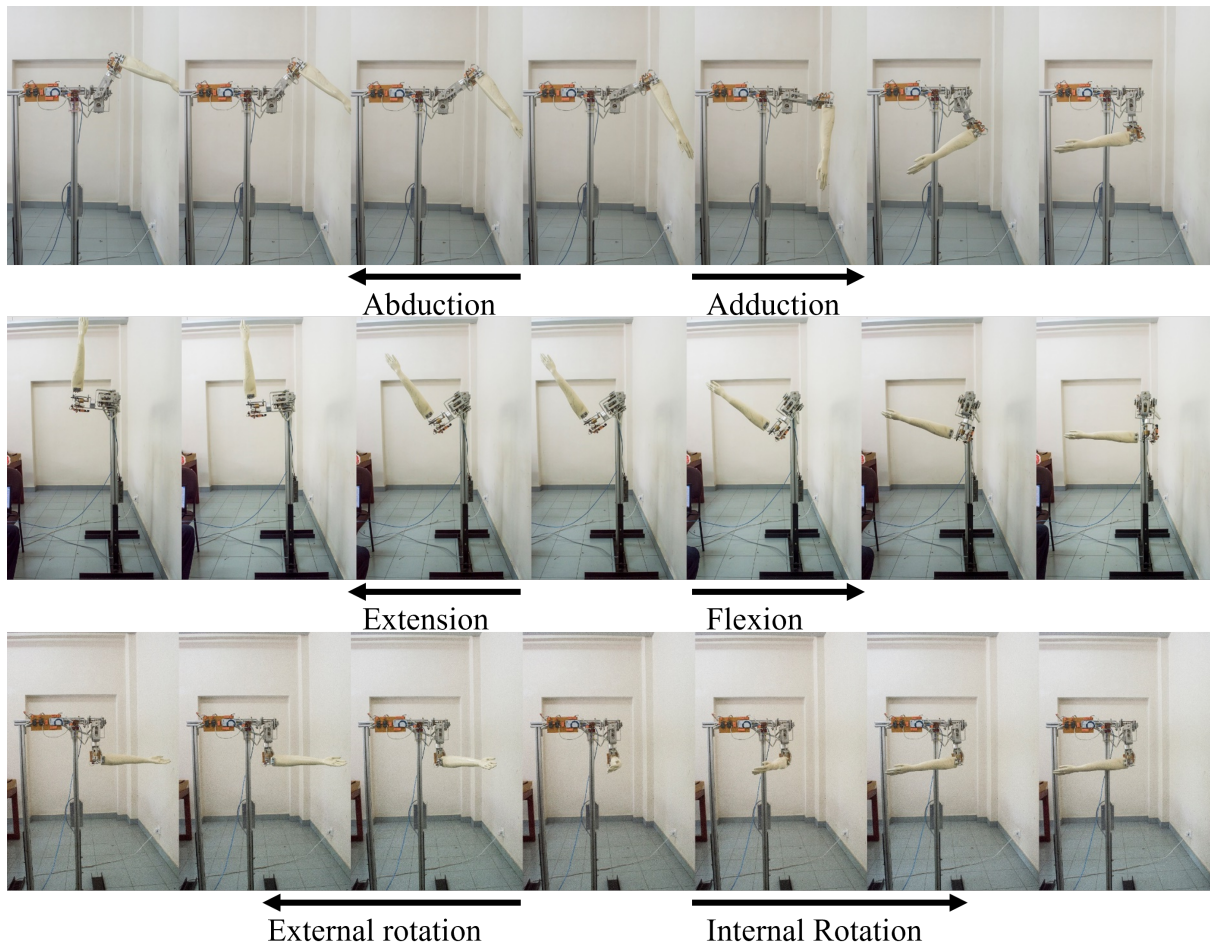


Figure 5.1. Motions of the shoulder prosthesis

Chapter 6 - Conclusion

In this thesis, a comprehensive design of an artificial active shoulder prosthetic device was presented. The device consists of 3 DOF, all active and powered by BLDC motors. The novel design presented in this thesis permitted the project group to overcome two of the notable issues with standalone shoulder prostheses that were developed; activating all three degrees of freedom, while maintaining the axes for abduction – adduction and flexion – extension at the same plane. This enables the prosthesis to function in a very similar way to an actual shoulder complex.

The range of motions of the prosthetic device are

- 0° - 150° for abduction – adduction
- (-20°) - 160° for flexion – extension
- 90° - (-90°) for internal rotation – external rotation

The motor control used both low level control (using PD control) and high level control (using surface EMG signals). The closed loop control system enabled the shoulder prosthesis to mimic the motion of the human shoulder once the EMG signals were acquired and transmitted.

6.1 Recommendations

The following can be suggested as future developments

- Reduce the overall size of the prosthesis, as the existing prosthesis can be fairly difficult for a shoulder amputee to use.
- Improve the EMG – based control system by utilizing feature extraction, to counteract the problems associated with surface EMG signals and artificial neural networks to better grasp the muscle signals and train the device accordingly.
- Implement the shoulder socket, with consideration to user comfort.

Chapter 7 - Appendices

7.1 Data obtained during each motion – RMS of muscle voltage signals (with the motion for which pronounced activity is observed)

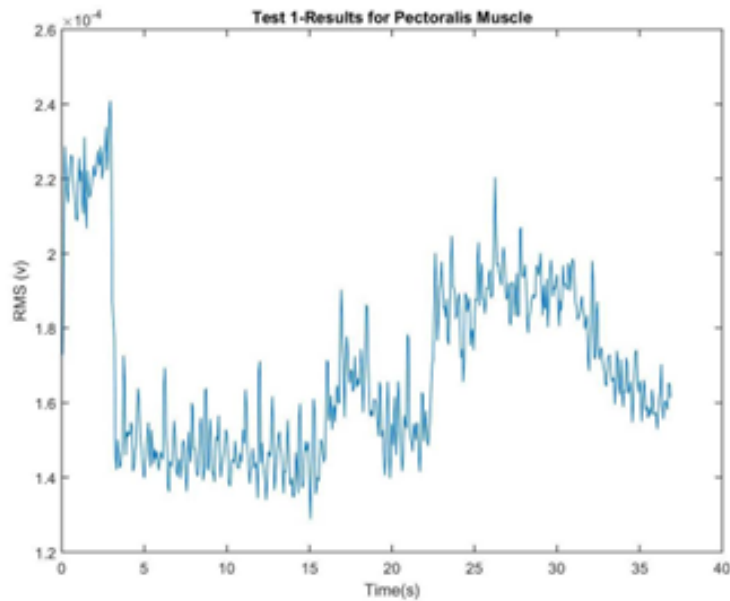


Figure 7.1. Results for pectoralis major muscle (flexion)

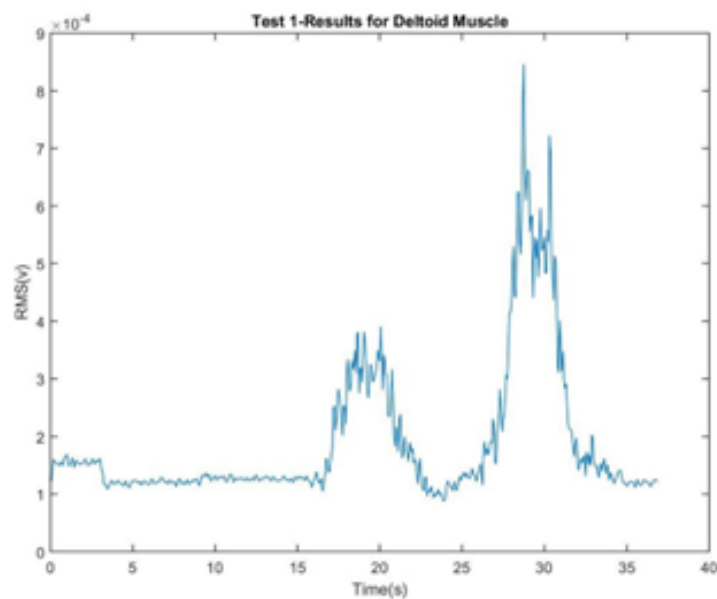


Figure 7.2. Results for anterior deltoid muscle (internal rotation)

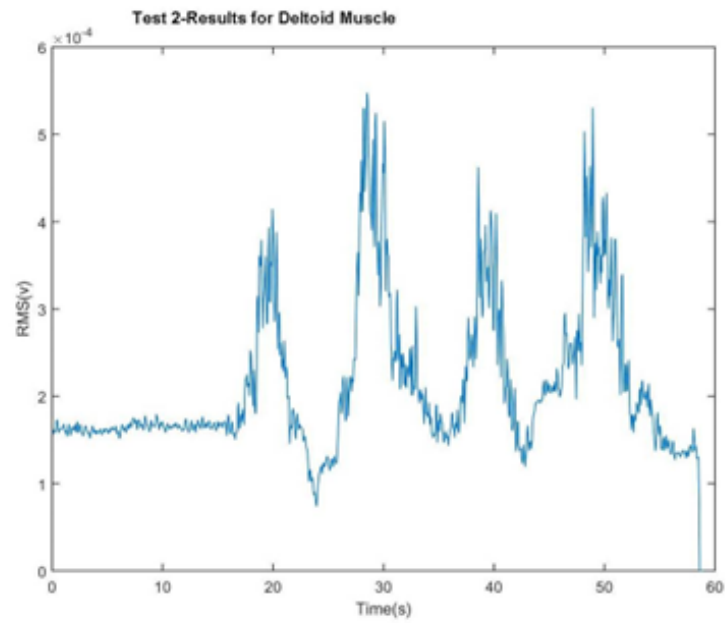


Figure 7.3. Results for lateral deltoid muscle (abduction)

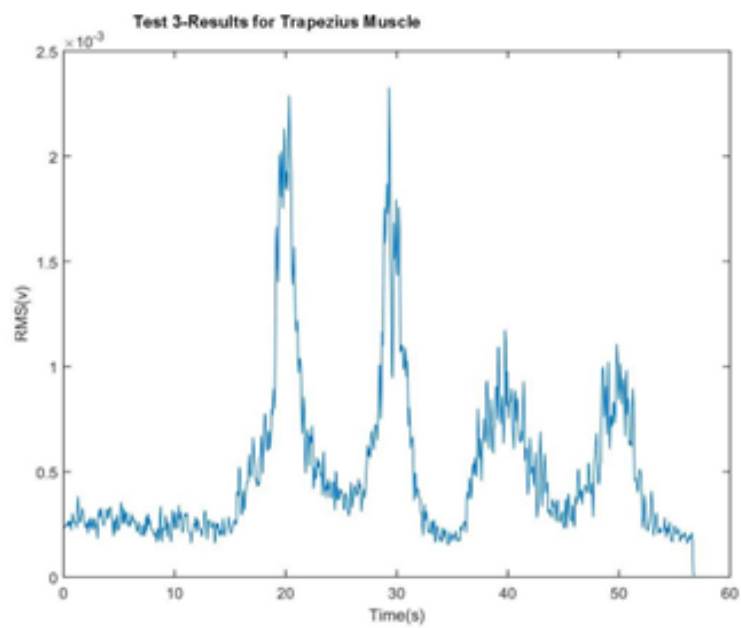


Figure 7.4. Results for Trapezius Muscle (adduction)

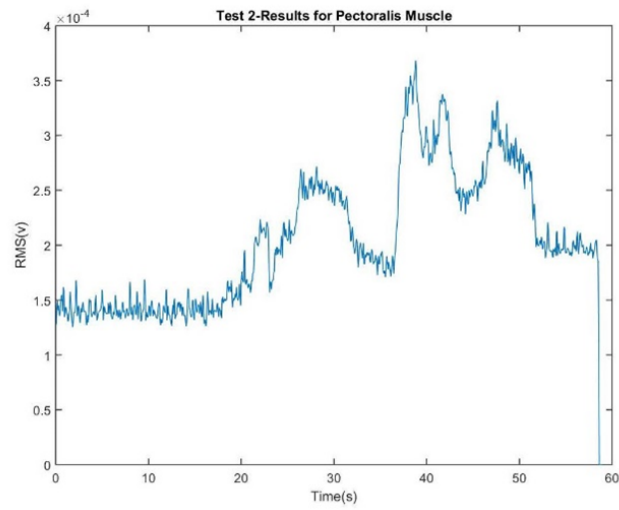


Figure 7.5. Results for pectoralis major muscle (flexion)

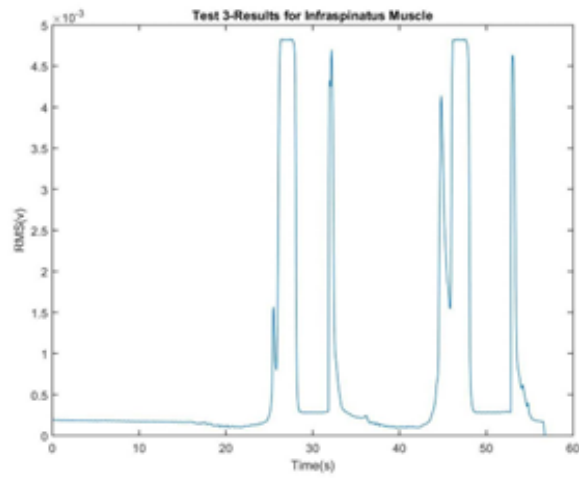


Figure 7.6. Results for Infraspinatus Muscle (external rotation)

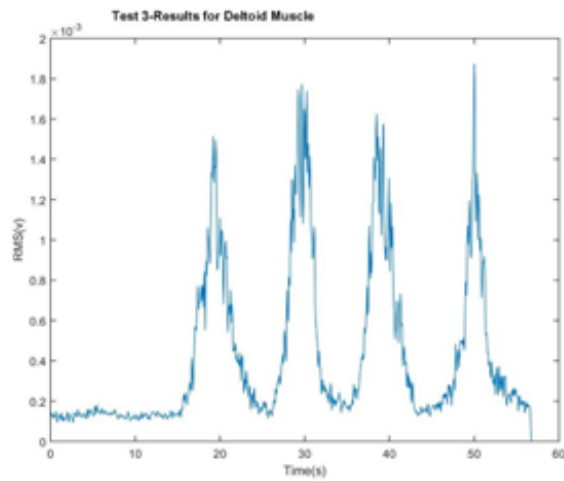


Figure 7.7. Results for posterior deltoid muscle (extension)

7.2 Program for the control of the prosthesis (Arduino and MATLAB)

```
1  Arduino Program
2  /* Group 7_FYP_Design & Development of Artificial Modular Shoulder Prosthesis
3     pwmFE    // upper motor (-) Flexion    // (+) Extension
4     pwmAbAd  // lower motor (-) Abduction // (+) Adduction*/
5  #define pwmPinFE 7          //10 or 7
6  #define enablePinFE 25      // 8 or 25
7  #define directionPinFE 23   // 9 or 23
8  #define interruptPinFE 2    // 2 or 3
9  #define pwmPinAbAd 10       //10 or 7
10 #define enablePinAbAd 8     // 8 or 25
11 #define directionPinAbAd 9   // 9 or 23
12 #define interruptPinAbAd 3   // 2 or 3
13 #define analogKp A0
14 #define analogKd A1
15 //////////////////////////////////////////////////Manual Control push buttons
16 #define m1Flexion 35
17 #define m1Extension 37
18 #define m2Abduction 39
19 #define m2Adduction 41
20 double pwmFE =0.0, pwmAbAd =0.0, pwmManual = 50;
21 int c=0;
22 //////////////////////////////////////////////////*****PID Tuning*****
23 int monitorInFE =0, monitorInAbAd =0, disFE =0,motorAngleFE =0;
24 int motorAngleAbAd =0, setValFE =0, setValAbAd =0;
25 const double kpFE =5, kdFE =1.2, kpAbAd =5, kdAbAd =1.2;
26 int valKpFE =0, valKdFE=0, valKpAbAd =0, valKdAbAd=0;
27
28 //////////////////////////////////////////////////*****Velocity measuring*****
29 long newPositionFE,newPositionFEAng, newPositionAbAd,newPositionAbAdAng ;
30 long oldPositionFE = 0,oldPositionAngFE =0, oldPositionAbAd = 0;
31 long oldPositionAngAbAd =0;
32 unsigned long newTimeFE,newAngFE, newTimeAbAd,newAngAbAd;
33 unsigned long oldTimeFE = 0 , oldAngFE =0, oldTimeAbAd = 0 , oldAngAbAd =0;
34 long velFE,desiredVelFE, velAbAd,desiredVelAbAd;
35
36 int serialreadvalue, pwmangFE, pwmangAbAd;
37 static boolean negative = false;
38 static boolean flexMotor = false;
39 static boolean abMotor = false;
40
41 void setup() {
42 //////////////////////////////////////////////////Motor 1 Flexion-Extension
43 pinMode(pwmPinFE,OUTPUT);
44 pinMode(enablePinFE,OUTPUT);
45 pinMode(directionPinFE,OUTPUT);
46 attachInterrupt(interruptPinFE, ISR1, RISING);
47 //////////////////////////////////////////////////Motor 2 Abduction-Adduction
48 pinMode(pwmPinAbAd,OUTPUT);
49 pinMode(enablePinAbAd,OUTPUT);
50 pinMode(directionPinAbAd,OUTPUT);
51 attachInterrupt(interruptPinAbAd, ISR2, RISING);
52 pinMode(13, OUTPUT); // Blink
53 //////////////////////////////////////////////////Manual Control push buttons
54 pinMode(m1Flexion, INPUT);
```

```

55     pinMode(m1Extension, INPUT);
56     pinMode(m2Abduction, INPUT);
57     pinMode(m2Adduction, INPUT);
58     Serial.begin(9600);
59 }
60 void loop() {
61     ////////////////////Blink
62     c++;
63     if (c<20) {
64         digitalWrite(13,HIGH);
65     }
66     else digitalWrite(13,LOW);
67     if ( c==1000) c=0;
68     ////////////////////*****Serial Read
69     if (Serial.available() > 0) {
70         serialreadvalue = 0;
71         while(Serial.available() > 0) {
72             char aChar = Serial.read();
73             if(aChar == 'f' || aChar == 'F'){
74                 flexMotor = true;
75                 abMotor = false; }
76             else if(aChar == 'a' || aChar == 'A'){
77                 flexMotor = false;
78                 abMotor = true; }
79             else if(aChar == '-') {
80                 negative = true;
81             }
82             else if(aChar == ' ' || aChar == '+'){
83                 negative = false; }
84             else if(aChar >= '0' && aChar <= '9') {
85                 serialreadvalue *= 10;
86                 serialreadvalue += aChar - '0';
87                 delay(100); } }
88             if(negative){
89                 serialreadvalue *= -1; } }
90     if (flexMotor){
91         pwmangFE = serialreadvalue; }
92     else if (abMotor){
93         pwmangAbAd = serialreadvalue; }
94     //////////////// Mapping motorIn value to motor angle///// 89*25
95     motorAngleFE=monitorInFE*2.0*360.0/2225.0;
96     //////////////// Velocity Measure
97     newpositionFE = monitorInFE;
98     newtimeFE = millis();
99     velFE = (newpositionFE-oldpositionFE) ;/* 1000 /(newtime-oldtime);
100     oldpositionFE = newpositionFE;
101     oldtimeFE = newtimeFE;
102     newpositionFEAng = pwmangFE;
103     newAngFE = millis();
104     desiredVelFE = (newpositionFEAng - oldpositionAngFE);
105     oldpositionAngFE = newpositionFEAng;
106     oldAngFE = newAngFE;
107     disFE = disFE + velFE ;
108     pwmFE = (((pwmangFE - motorAngleFE)*kpFE)+ (desiredVelFE - velFE)*kdFE )*256/360;
109     pwmAbAd = ((pwmangAbAd - motorAngleAbAd)*kpAbAd)+ (desiredVelAbAd - velAbAd)*kdFE;
110     pwmAbAd = pwmAbAd*256/360;
111     //pwmFE = pwmangFE;

```

```

112 //pwmAbAd = pwmangAbAd;
113 // Flexion - Extension Positioning with Direction pin and Monitor Pin
114 if(pwmFE>0.1 && pwmFE<256){
115     digitalWrite(enablePinFE,HIGH);
116     analogWrite(pwmPinFE,pwmFE);
117     digitalWrite(directionPinFE,HIGH);
118     setValFE=0; }
119 else if(pwmFE<-0.1 && pwmFE>-256){
120     digitalWrite(enablePinFE,HIGH);
121     analogWrite(pwmPinFE,-pwmFE);
122     digitalWrite(directionPinFE,LOW);
123     setValFE=1; }
124 else if(pwmFE==0){
125     digitalWrite(enablePinFE,LOW);
126     analogWrite(pwmPinFE,-pwmFE);
127     digitalWrite(directionPinFE,LOW); }
128 // Abduction - Adduction Positioning with Direction pin and Monitor Pin
129 if(pwmAbAd>0.1 && pwmAbAd<256){
130     digitalWrite(enablePinAbAd,HIGH);
131     analogWrite(pwmPinAbAd,pwmAbAd);
132     digitalWrite(directionPinAbAd,HIGH);
133     setValAbAd=0; }
134 else if(pwmAbAd<-0.1 && pwmAbAd>-256){
135     digitalWrite(enablePinAbAd,HIGH);
136     analogWrite(pwmPinAbAd,-pwmAbAd);
137     digitalWrite(directionPinAbAd,LOW);
138     setValAbAd=1; }
139 else if(pwmAbAd==0){
140     digitalWrite(enablePinAbAd,LOW);
141     analogWrite(pwmPinAbAd,-pwmAbAd);
142     digitalWrite(directionPinAbAd,LOW);
143 }
144 /////////////////////////////////////////////////////////////////// Interrupt Pin Function
145 void ISR1() {
146     if(setValFE==0) monitorInFE++;
147     else if(setValFE==1) monitorInFE--;
148 }
149 void ISR2() {
150     if(setValAbAd==0) monitorInAbAd++;
151     else if(setValAbAd==1) monitorInAbAd--;
152 }

```

```

1  %% MATLAB Program
2  t =0; mtmap =0;
3  % EMG setup
4  emg = daq.createSession('ni');
5  ch1 = emg.addAnalogInputChannel('Dev1', 4, 'Voltage');
6  ch2 = emg.addAnalogInputChannel('Dev1', 5, 'Voltage');
7
8  emg.Rate = 2000; % sampling rate
9  lh = emg.addlistener('DataAvailable',@stopWhenExceed);
10
11 % data per one feature set (total of all channels)
12 emg.NotifyWhenDataAvailableExceeds = 100;
13 emg.IsContinuous = true;
14 emg.startBackground();
15
16 %serial open
17 avail = instrfindall('status','open');
18 if ~isempty(avail) fclose(avail);
19 end
20
21 ardFE = serial('COM7', 'BaudRate', 9600);
22 fopen(ardFE);
23 disp('here');
24 pause(2);
25 disp('here2');
26 % fileID = fopen('EMGdataWithShoulder.txt','w');
27 %open file
28 % fprintf(fileID,'%6s %12s\n','T value');
29 while (1)
30     pause(0.01);
31     if (getGlobalSx==0)
32         data = getGlobalx;
33         setGlobalSx(1);
34     end
35     emgRMS = rms(data);
36     fl = emgRMS(1);
37     ex = emgRMS(2);
38     mt = (100*ex-110*fl);
39     mt = mt +65;
40     if mtmap < 0 mtmap = 0;
41     elseif mt > 128 mt = 0;
42     end
43     disp(mt)
44     sv = char(mt);
45     fprintf(ardFE,'%c',sv);
46 end

```

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